NAVAL STATION NORFOLK, VIRGINIA

Q-AREA FISHING PIER

NAVY PUBLIC WORKS CENTER
DESIGN DIVISION

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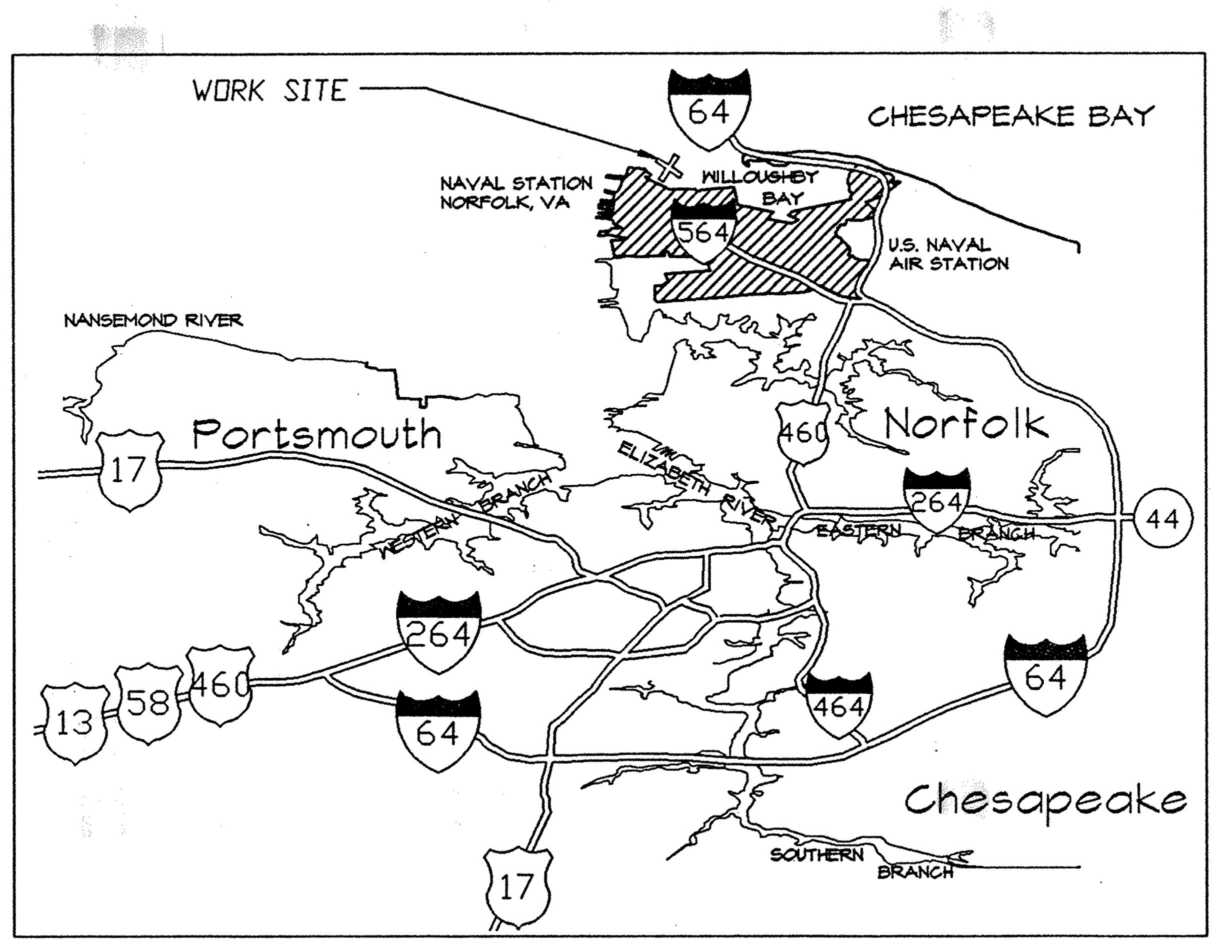
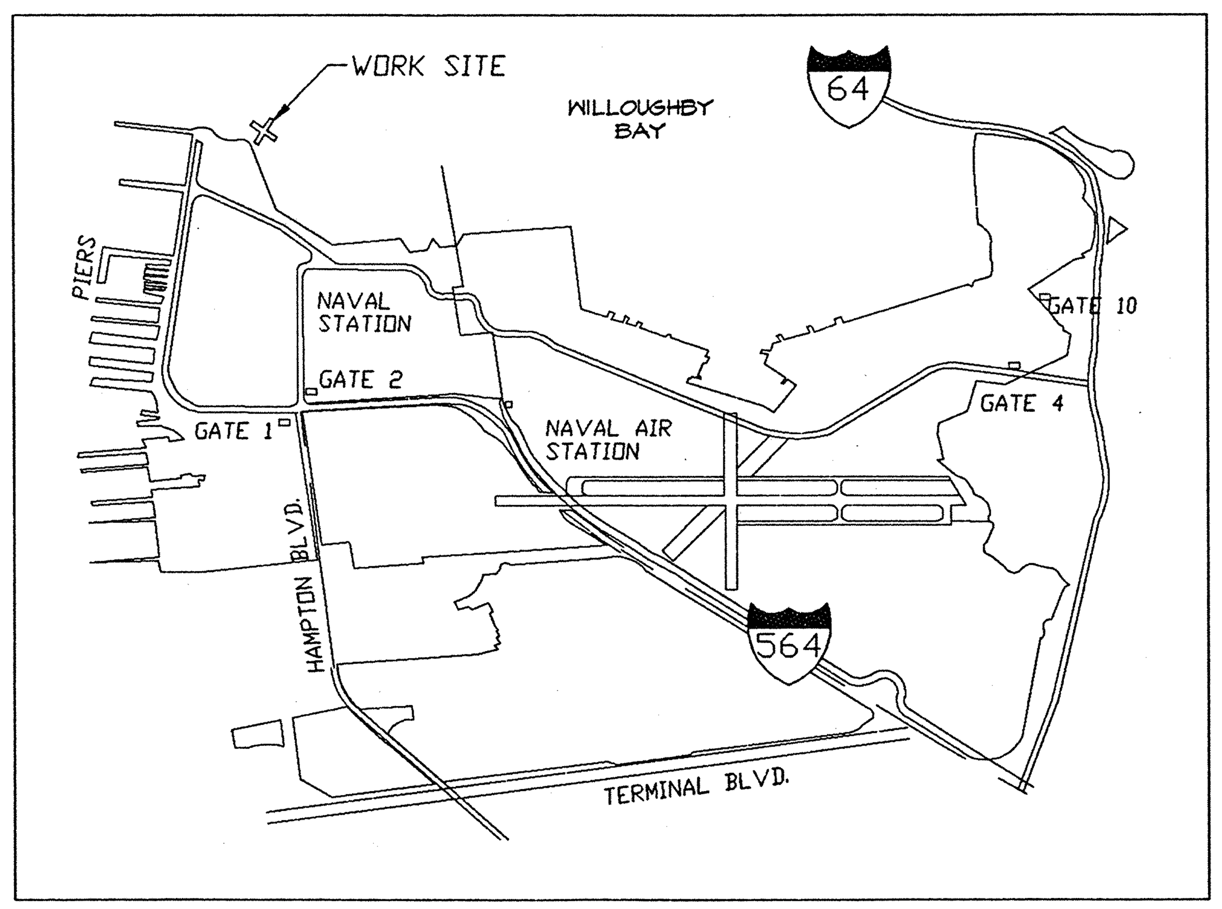


VICINITY PLAN

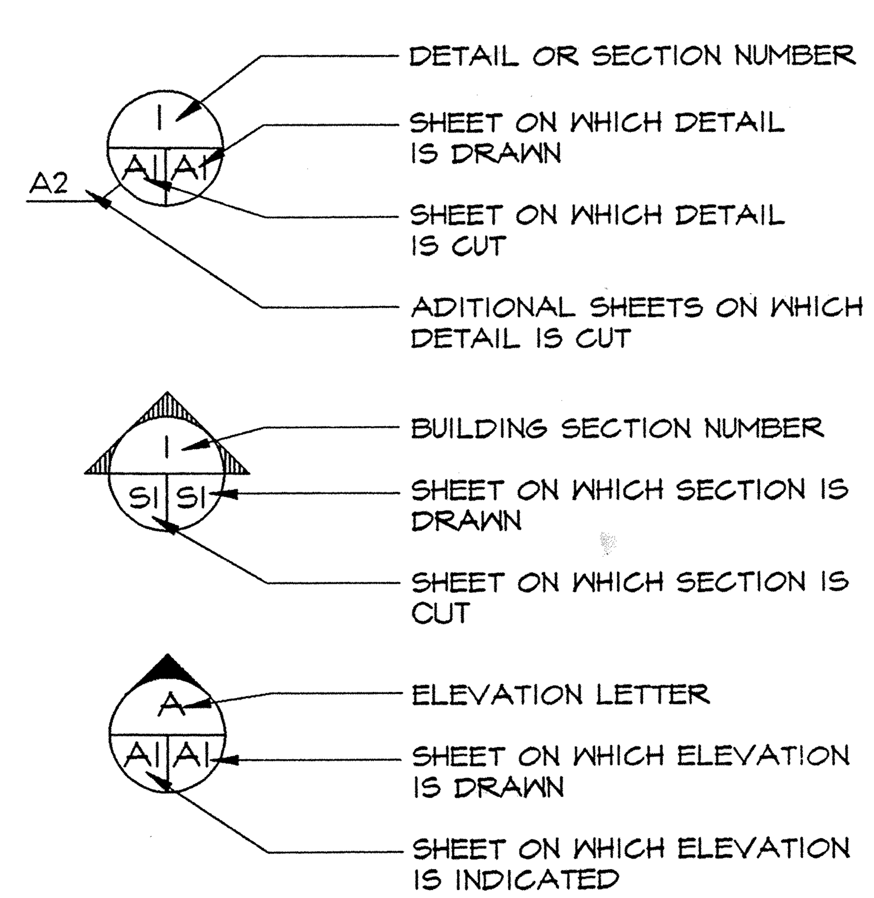
NOT TO SCALE



LOCATION PLAN

SCALE: FULL SIZE

SYMBOLS LEGEND



7/10/97 AS BUILT

SYM | DESCRIPTION | DATE | APPROVED

REVISIONS

SUBMITTED BY: FIRM MEMBER / PRINCIPAL

APPROVED: DATE: SANDRA HAWKINSON 06/03/96

SATISFACTORY TO ACTIVITY

APPROVED: DATE:

SATISFACTORY TO COMNAVBASE

JOB ORDER NO. 1646544

DES. EJA DR. EJA CHK. MKM

E.I.C. MKM

BR. MGR. F. B. COLE, JR. P.E.

DIV. DIR. R. H. IVES, JR. P.E.

APPROVED: DATE:

FOR C.O. COMMANDER, NAVFAC

PWC NORVA

SEAL

DEPT. OF THE NAVY NAVFACENGCOM

NAVY PUBLIC WORKS CENTER NORFOLK, VIRGINIA

NAVAL STATION NORFOLK, VIRGINIA

FISHING PIER Q-AREA

TITLE SHEET

CODE ID. NO. 80091

DRAWING SIZE: D

CONST. CONT. NO. N62470-96-B-7429

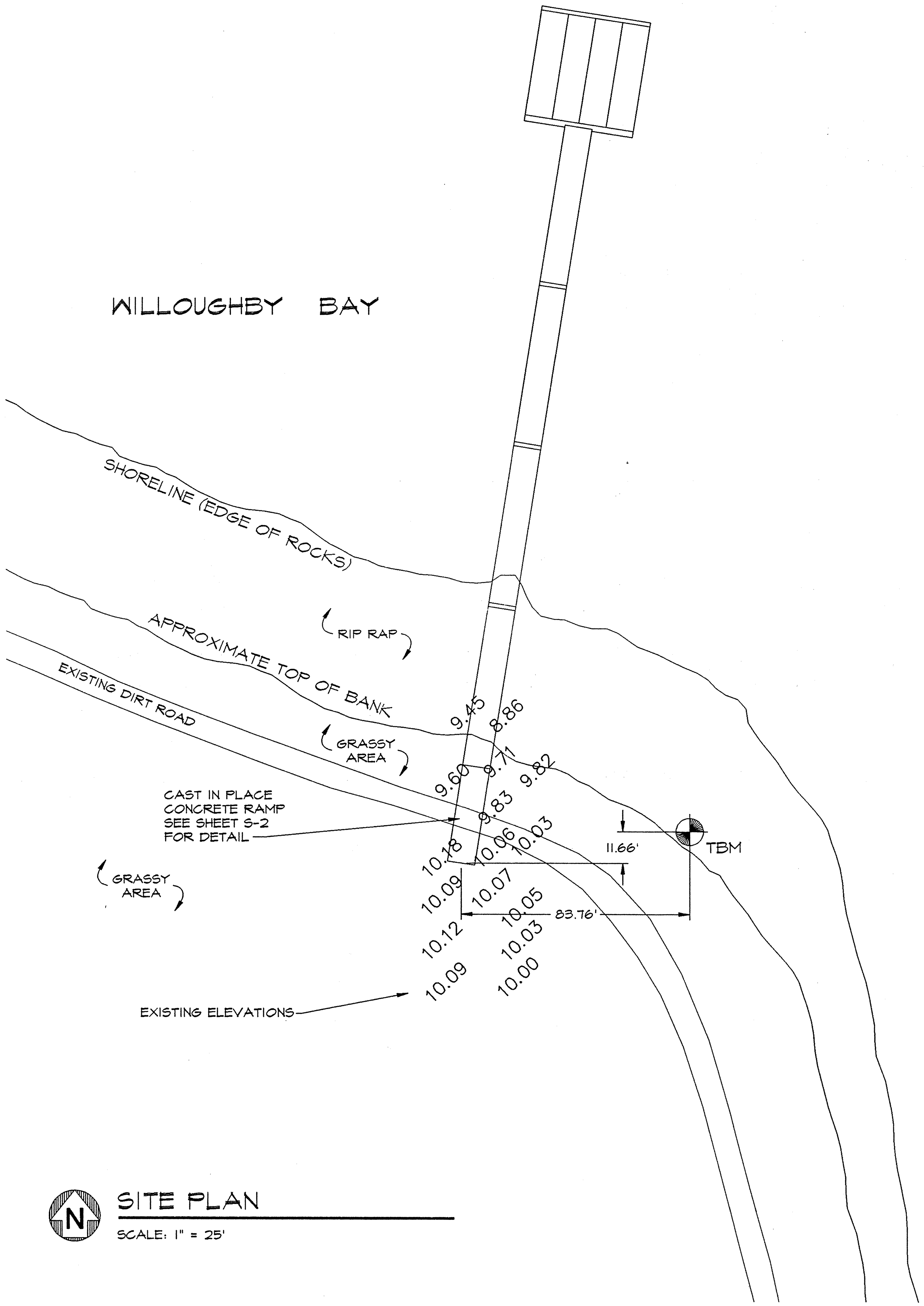
SPEC. 05-96-7429

NAVFAC DRAWING NO. 4,333,335

PWC DRAWING NO. 15,499

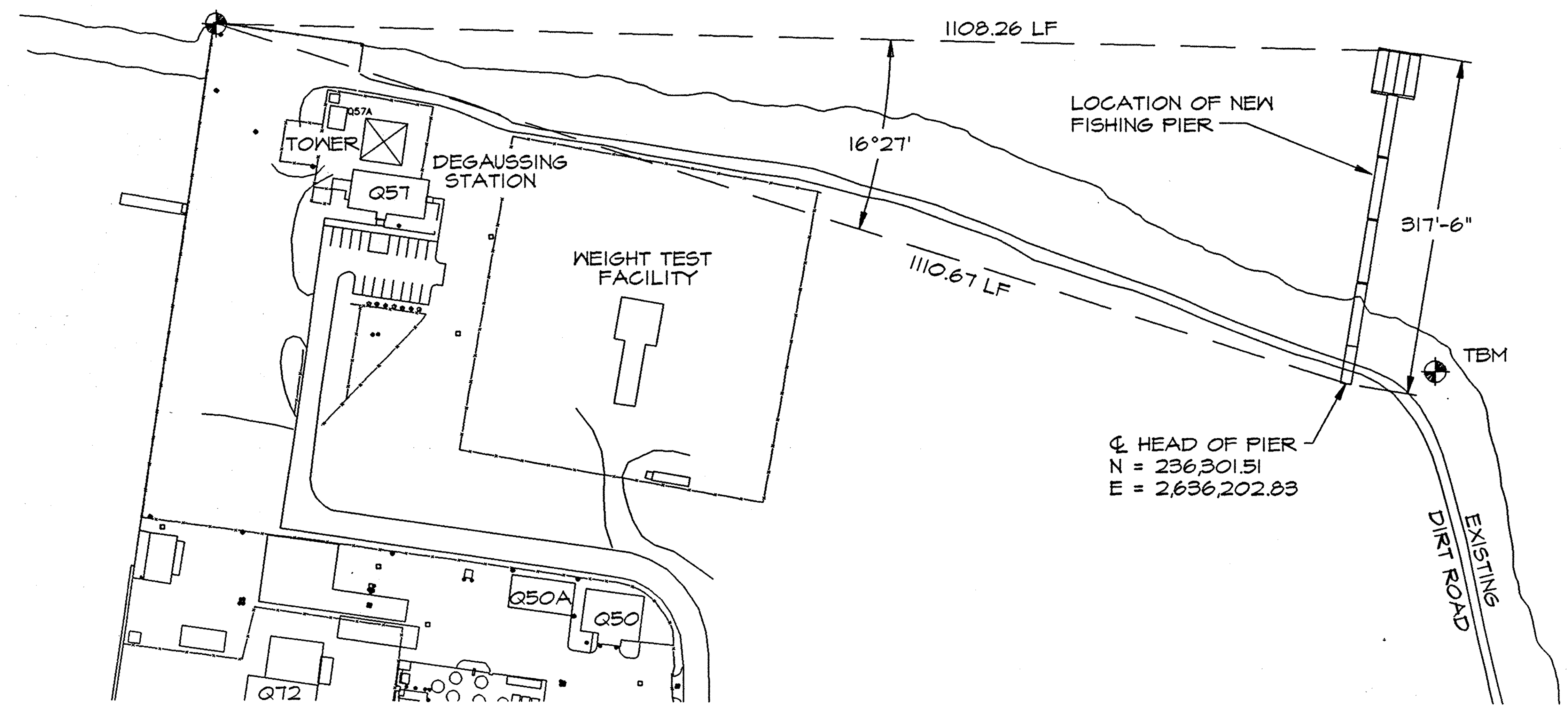
SHEET 1 OF 13

T-1



SITE PLAN

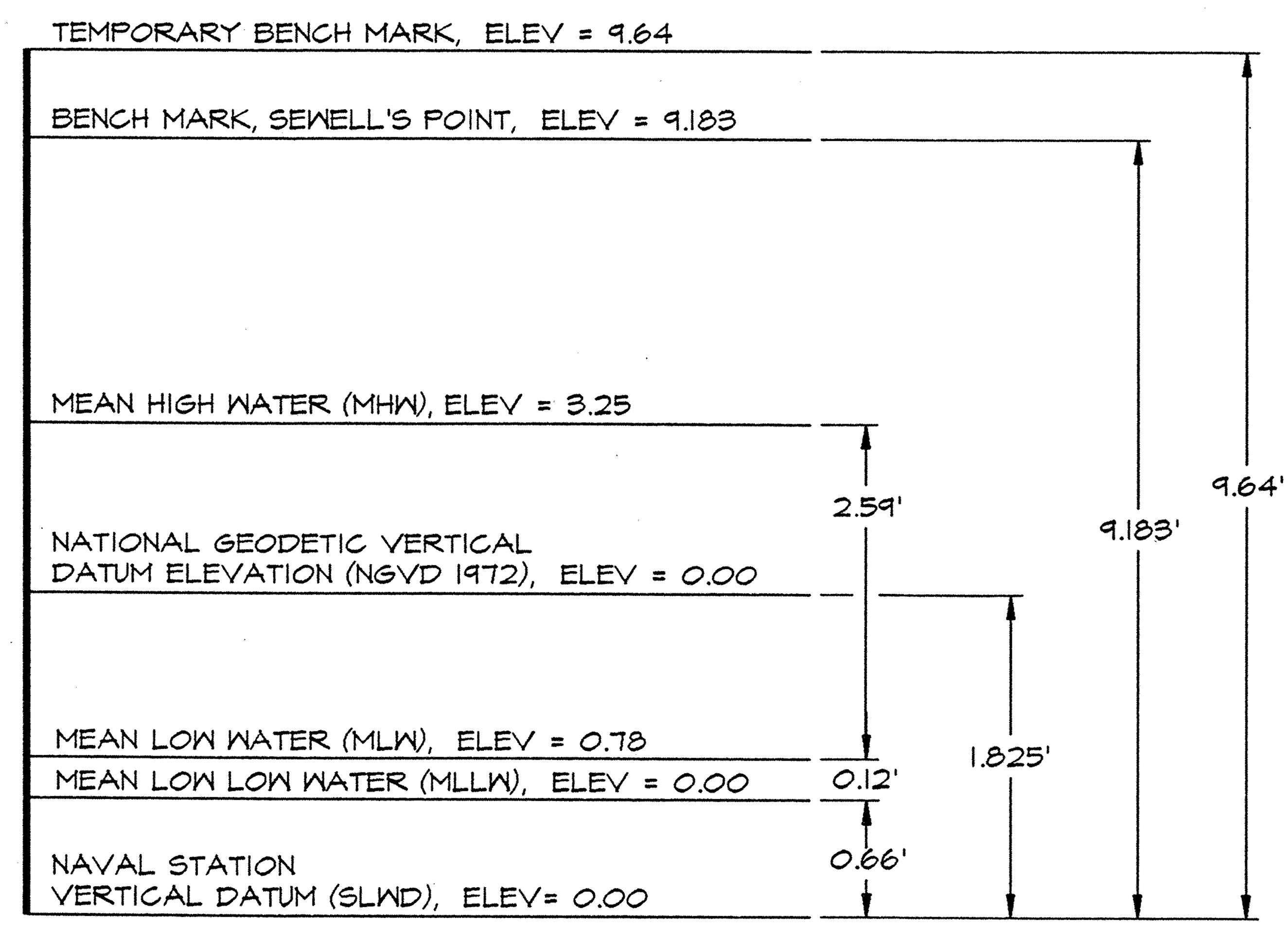
SCALE: 1" = 25'



LOCATION PLAN

SCALE: 1" = 100'

SURVEY DATUM INFORMATION

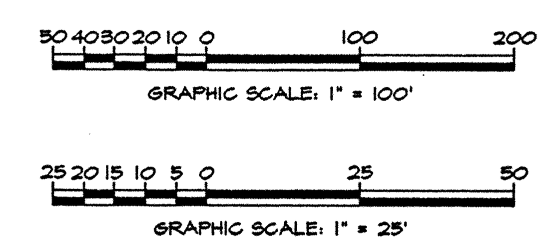


GENERAL NOTES

1. TEMPORARY BENCH MARK, TBM, IS A PAINTED SQUARE ON TOP OF A CONCRETE HEADWALL AT THE TOP OF THE BANK AND HAS AN ELEVATION OF 9.64 AT N = 236,313.17 AND E = 2,636,286.59.

2. TOPOGRAPHIC SURVEY WAS COMPLETED BY PWC ENGINEERING IN MAY 1996. ALL ELEVATIONS ARE IN FEET AND REFER TO NAVAL STATION DATUM UNLESS OTHERWISE NOTED. DATUM IS BASED ON 0.00 = STATION LOW WATER, WHICH IS 1.825 FEET BELOW NGVD OF 1929 (1972 ADJUSTMENT).

GRAPHIC SCALES



SYM	DESCRIPTION	DATE	APPROVED
	AS BUILT	7/10/97	
	REVISIONS		

JOB ORDER NO. 1646544
DES. CLW DR. CLW CHK.
E.I.C. M. MONTGOMERY
BR. MGR. F. B. COLE, JR. P.E.
DIV. DIR R. H. IVES, JR. P.E.
APPROVED: DATE:
FOR C.O. COMMANDER, NAVFAC

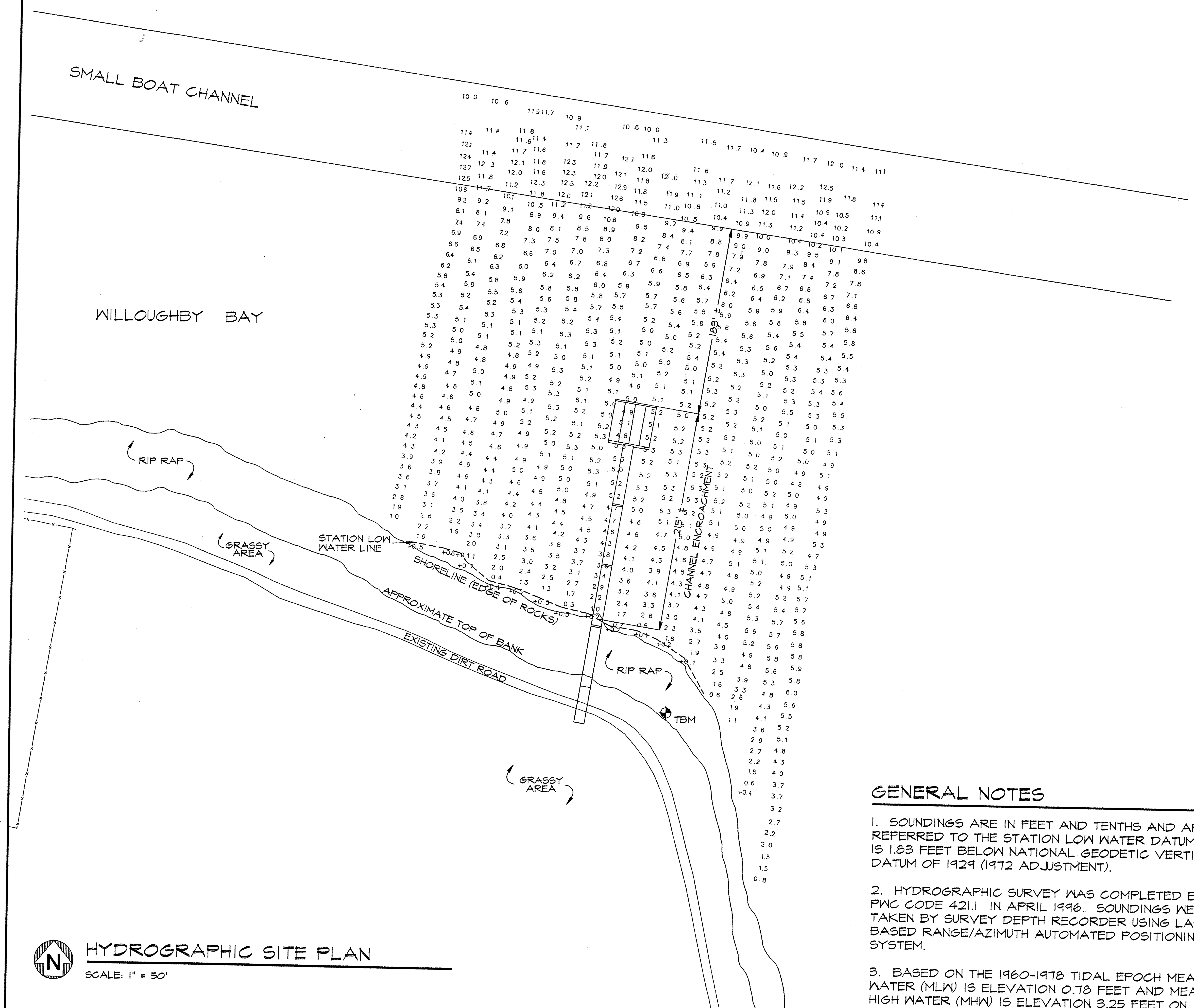
SUBMITTED BY:
FIRM MEMBER / PRINCIPAL
APPROVED: Sandra L.Hawkinson 3 June 1996 DATE:
SATISFACTORY TO ACTIVITY
APPROVED: DATE:
SATISFACTORY TO COMNAVBASE

P W C NORVA
SEAL

DEPT. OF THE NAVY NAVFACENGCOM
NAVY PUBLIC WORKS CENTER
NAVAL STATION NORFOLK, VIRGINIA
NAVAL STATION NORFOLK, VIRGINIA
FISHING PIER
Q-AREA
SITE PLAN

CODE ID. NO. 80091
DRAWING SIZE: D
CONST. CONT. NO. N62470-96-B-7429
SPEC. 05-7429
NAVFAC DRAWING NO. 4,333,336
PWC DRAWING NO. 15,499 A

C-1



HYDROGRAPHIC SITE PLAN

SCALE: 1" = 50'

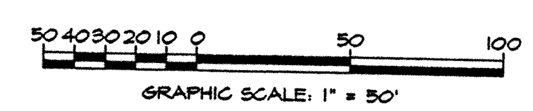
GENERAL NOTES

1. SOUNDINGS ARE IN FEET AND TENTHS AND ARE REFERRED TO THE STATION LOW WATER DATUM WHICH IS 1.83 FEET BELOW NATIONAL GEODETIC VERTICAL DATUM OF 1929 (1972 ADJUSTMENT).

2. HYDROGRAPHIC SURVEY WAS COMPLETED BY PWC CODE 421.1 IN APRIL 1996. SOUNDINGS WERE TAKEN BY SURVEY DEPTH RECORDER USING LASER BASED RANGE/AZIMUTH AUTOMATED POSITIONING SYSTEM.

3. BASED ON THE 1960-1978 TIDAL EPOCH MEAN LOW WATER (MLW) IS ELEVATION 0.78 FEET AND MEAN HIGH WATER (MHW) IS ELEVATION 3.25 FEET ON STATION LOW WATER DATUM (SLWD).

GRAPHIC SCALE



SYM	DESCRIPTION	DATE	APPROVED
	AS BUILT	7/10/97	FBCole

REVISIONS

SUBMITTED BY:
FIRM MEMBER / PRINCIPAL
APPROVED: DATE:
Sandra L.Hawkinson 3 June 1996
SATISFACTORY TO ACTIVITY
APPROVED: DATE:
SATISFACTORY TO COMNAVBASE

JOB ORDER NO. 1646544
DES. BMS | DR. CLW | CHK JWB
E.I.C. M. MONTGOMERY
BR. MGR. F. B. COLE, JR. P.E.
DIV. DIR R. H. IVES, JR. P.E.
APPROVED: F. B. Cole DATE: 6/27/96
FOR C.O. COMMANDER, NAVFAC

P W C NORVA SEAL

DEPT. OF THE NAVY NAVFACENGCOM
NAVY PUBLIC WORKS CENTER
NAVAL STATION NORFOLK, VIRGINIA
NAVAL STATION NORFOLK, VIRGINIA
FISHING PIER
Q-AREA
HYDROGRAPHIC SITE PLAN

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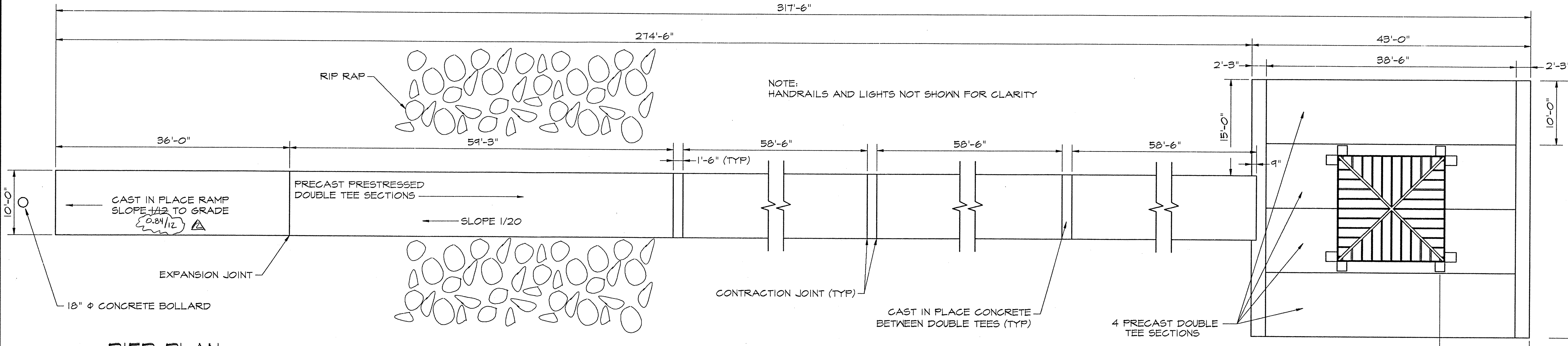
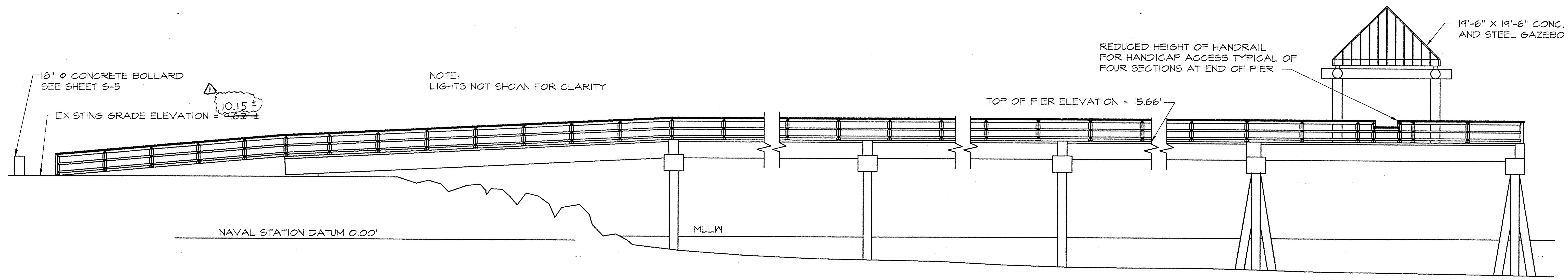
SPEC. 05-7429
NAVFAC DRAWING NO. 4,333,337
PWC DRAWING NO. 15,499 B
SHEET 3 OF 13
C-2



PIER PLAN

SCALE: 1/8" = 1'-0"



PIER PROFILE

SCALE: 1/8" = 1'-0"

GENERAL NOTES

1. THE SCOPE OF WORK INCLUDES BUT IS NOT LIMITED TO CONSTRUCTING A CONCRETE FISHING PIER CONSISTING OF A 274'-6" LONG APPROACH LEADING TO A 43' X 40' PLATFORM, A CONCRETE FRAMED GAZEBO AND LIGHTS.

2. ALL PILES SHALL BE PRE-CAST PRESTRESSED CONCRETE DESIGNED IN ACCORDANCE WITH P.C.I. WITH A MINIMUM MOMENT CAPACITY OF 140 FT-KIPS CONCURRENT WITH A 100 KIP (50 TON) AXIAL LOAD OR 50 KIP (25 TON) TENSILE LOAD. ALL PILES SHALL BE BATTERED 2/12 IN THE DIRECTION SHOWN. FOR BIDDING PURPOSES PRODUCTION PILES SHALL BE 84' LONG, TWO TEST PILES INDICATED AS (■) SHALL EACH BE 94' LONG. UNLESS OTHERWISE NOTED TIP ELEVATION FOR ALL PILES SHALL BE -72.17 AND CUT OFF ELEVATION SHALL BE 11.83. MAXIMUM CUT OFF LENGTH SHALL BE 3' FOR PRODUCTION PILES AND 13' FOR TEST PILES.

3. PRECAST PRESTRESSED DOUBLE TEE DECK SECTIONS SHALL BE DESIGNED IN ACCORDANCE WITH P.C.I. FOR THE FOLLOWING LOADS:

LIVE LOAD	100 PSF APPROACH 175 PSF 43' X 40' END OF PIER
LATERAL LOAD	350 PLF
GAZEBO POINT LOADS	8,500 lbs DEAD AND 500 lbs LIVE LOAD

4. ALL CONCRETE WORK SHALL BE IN ACCORDANCE WITH ACI 318-95.

5. ALL PRECAST AND CAST IN PLACE CONCRETE SHALL HAVE A MINIMUM COMPRESSIVE STRENGTH OF 5,000 PSI AT 28 DAYS, AND SHALL HAVE 5-7% ENTRAINED AIR, MAXIMUM WATER CEMENT RATIO OF 0.40, 10% OF TOTAL CEMENTIOUS MATERIAL SHALL BE SILICA FUME. SPECIAL PRECAUTIONS SHALL BE TAKEN TO ENSURE CORRECT CURING PROCEEDURES ARE FOLLOWED.

6. ALL REINFORCING STEEL SHALL BE MADE CONTINUOUS BY LAPPING BARS A MINIMUM OF 40 BAR DIAMETERS BUT NOT LESS THAN 2'-0". REINFORCING STEEL SHALL CONFORM TO ASTM A615, GRADE 60, EPOXY COATED.

7. MINIMUM CONCRETE COVER FOR REINFORCING SHALL BE 3" FOR ALL CAST IN PLACE AND 1½" FOR PRECAST CONCRETE UNLESS OTHERWISE NOTED.

8. ALL TIMBER SHALL BE NUMBER 1 GRADE SOUTHERN YELLOW PINE TREATED TO 0.40 PCF RETENTION OF CCA.

9. ALL STRUCTURAL STEEL HANDRAIL COMPONENTS SHALL CONFORM TO ASTM A-36 AND BE HOT DIPPED GALVANIZED. ALL STEEL TUBE SECTIONS SHALL CONFORM TO ASTM A-500 GRADE B AND SHALL BE SHOP PRIMED AND PAINTED.

10. ALL HANDRAIL BOLTS, NUTS, AND WASHERS SHALL BE GALVANIZED. PAINT THREADS OF ALL GALVANIZED BOLTS AFTER INSTALLATION USING ASTM A 780 ZINC RICH PAINT.

11. ALL WELDING SHALL BE IN ACCORDANCE WITH AWS D1.1 USING E70XX ELECTRODES.

12. COORDINATE WITH MANUFACTURER OF PRECAST DOUBLE TEE SECTIONS LOCATION OF HOLES FOR MOUNTING HANDRAILS PRIOR TO DRILLING. DO NOT CUT OR EXPOSE ANY STEEL IN THE DOUBLE TEES WITHOUT WRITTEN APPROVAL FROM THE MANUFACTURER.

13. APPLY LINSEED OIL EMULSION TO SURFACE OF CONCRETE DECK AND ALL TIMBER SURFACES.

14. ALL ELEVATIONS ARE REFERENCED TO NAVAL STATION VERTICAL DATUM ELEVATION = 0.00'. ELEVATION OF MEAN LOW WATER = 0.66' REFERENCED TO NAVAL STATION DATUM.

15. CONTRACTOR SHALL VERIFY WATER DEPTHS PRIOR TO MOBILIZING BARGE FOR PILE DRIVING OPERATIONS.

16. THERE SHALL BE NO VEHICULAR TRAFFIC ON THE PIER AT ANY TIME.

17. METAL ROOF AND CONNECTORS SHALL BE DESIGNED FOR 50 PSF DOWNWARD AND UPLIFT PRESSURE.

18. EXPOSED ANCHOR BOLTS FOR THE GAZEBO SHALL BE FIELD PAINTED TO MATCH THE STEEL STRUCTURE.

19. EXPOSED ENDS OF ROOF FASTENERS SHALL BE FIELD PAINTED TO MATCH THE STEEL STRUCTURE.

20. THE BASE BID INCLUDES THE PIER, CONDUIT EMBEDDED IN THE PIER AND THE NAVIGATIONAL OBSTRUCTION LIGHTING.

21. ADDITIVE BID ITEM NO. 1 INCLUDES THE GAZEBO.

22. ADDITIVE BID ITEM NO. 2 INCLUDES PIER AND GAZEBO LIGHTING, ELECTRIFICATION AND ASSOCIATED WORK.
BID ITEM NO. 2 NOT IN CONTRACT

GRAPHIC SCALE

1/8" = 1'-0" 5' 0 5' 10' 20'

SYM	DESCRIPTION	DATE	APPROVED
	AS BUILT	7/10/97	

REVISIONS

JOB ORDER NO. 1646544
DES. EJA | DR. EJA | CHK. MKM
E.I.C. MKM
BR. MGR. F. B. COLE, JR. P.E.
DIV. DIR. R. H. IVES, JR. P.E.
APPROVED: DATE: 6/21/96
FOR C.O. COMMANDER, NAVFAC

SUBMITTED BY:
FIRM MEMBER / PRINCIPAL
APPROVED: DATE:
SANDRA HAWKINSON 06/03/96
SATISFACTORY TO ACTIVITY
APPROVED: DATE:
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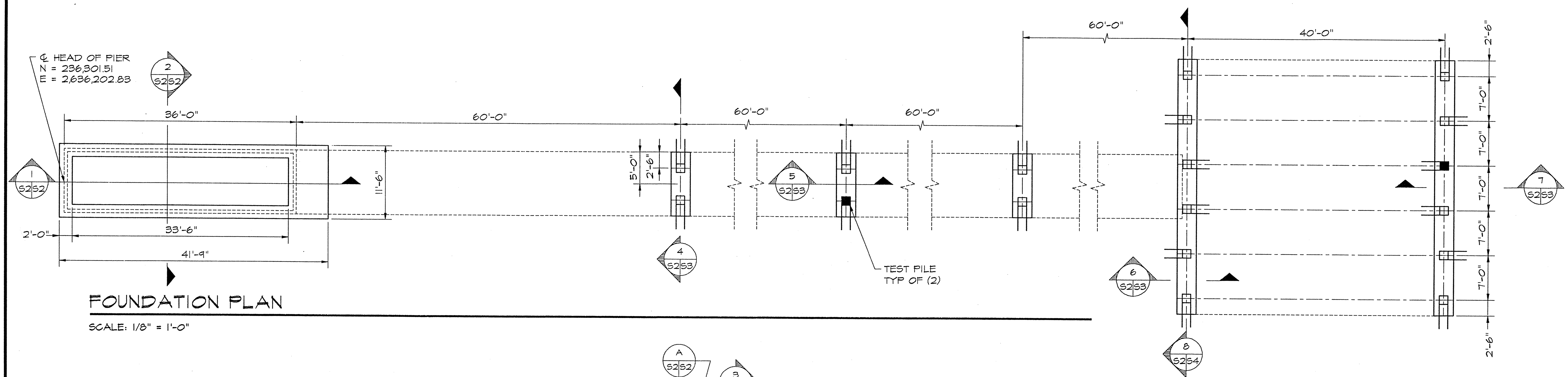
P W C
NORVA
SEAL

DEPT. OF THE NAVY NAVFACENGCOM
NAVY PUBLIC WORKS CENTER
NAVAL STATION NORFOLK, VIRGINIA
NAVAL STATION NORFOLK, VIRGINIA
FISHING PIER
Q-AREA
PIER PLAN

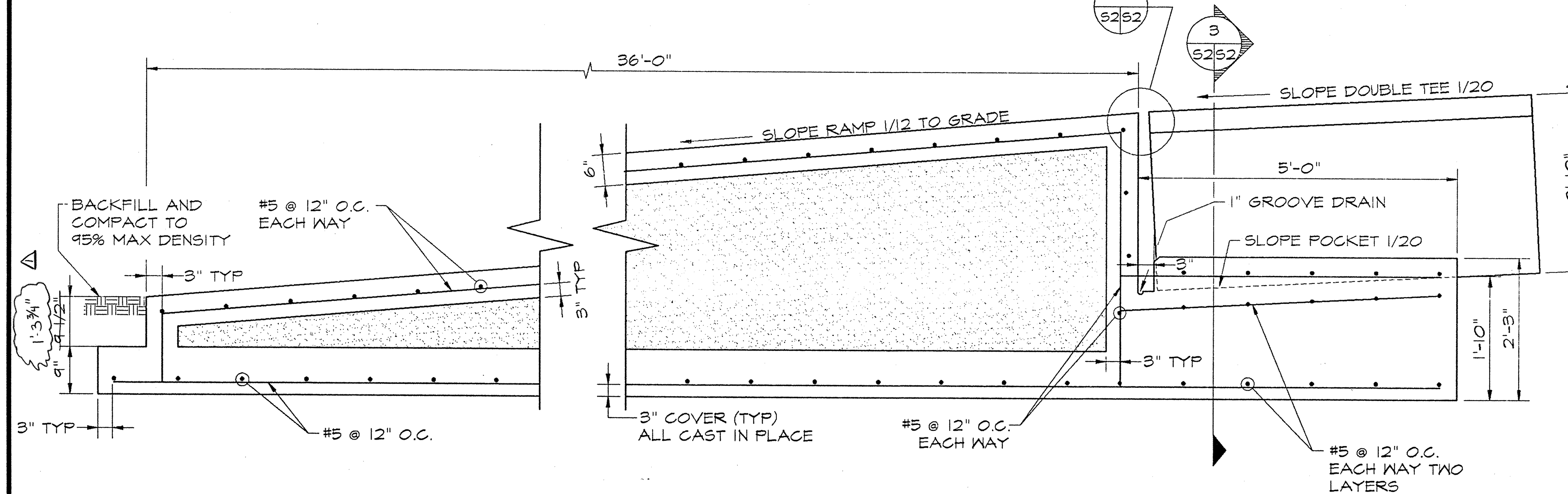
CODE ID. NO. 80091
DRAWING SIZE: D
CONST. CONT. NO. N62470-96-B-7429
SPEC. 05-96-7429
NAVFAC DRAWING NO. 4333338
PWC DRAWING NO. 15,499 C
SHEET 4 OF 13

S-1



FOUNDATION PLAN

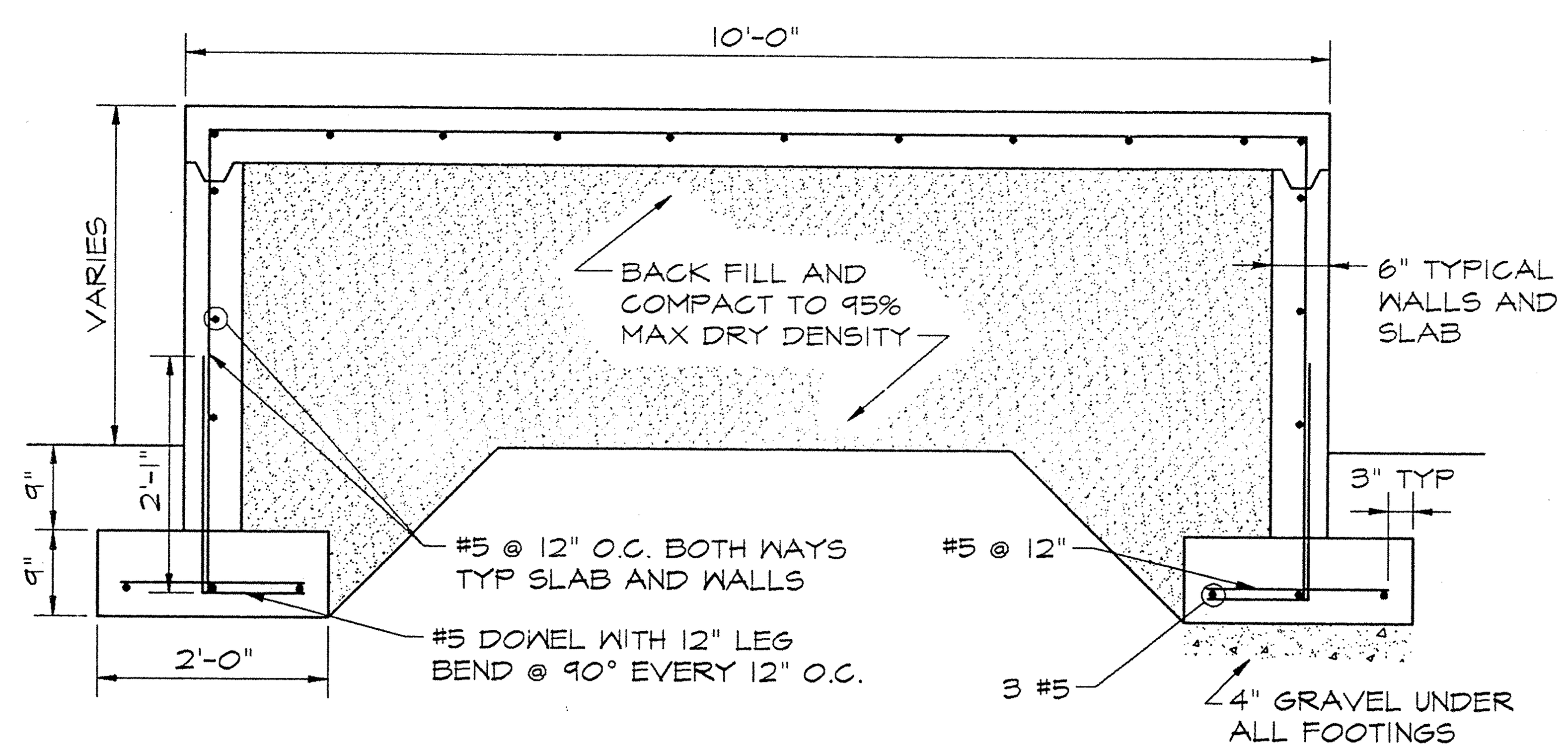
SCALE: 1/8" = 1'-0"



RAMP SECTION

1 S2 S2

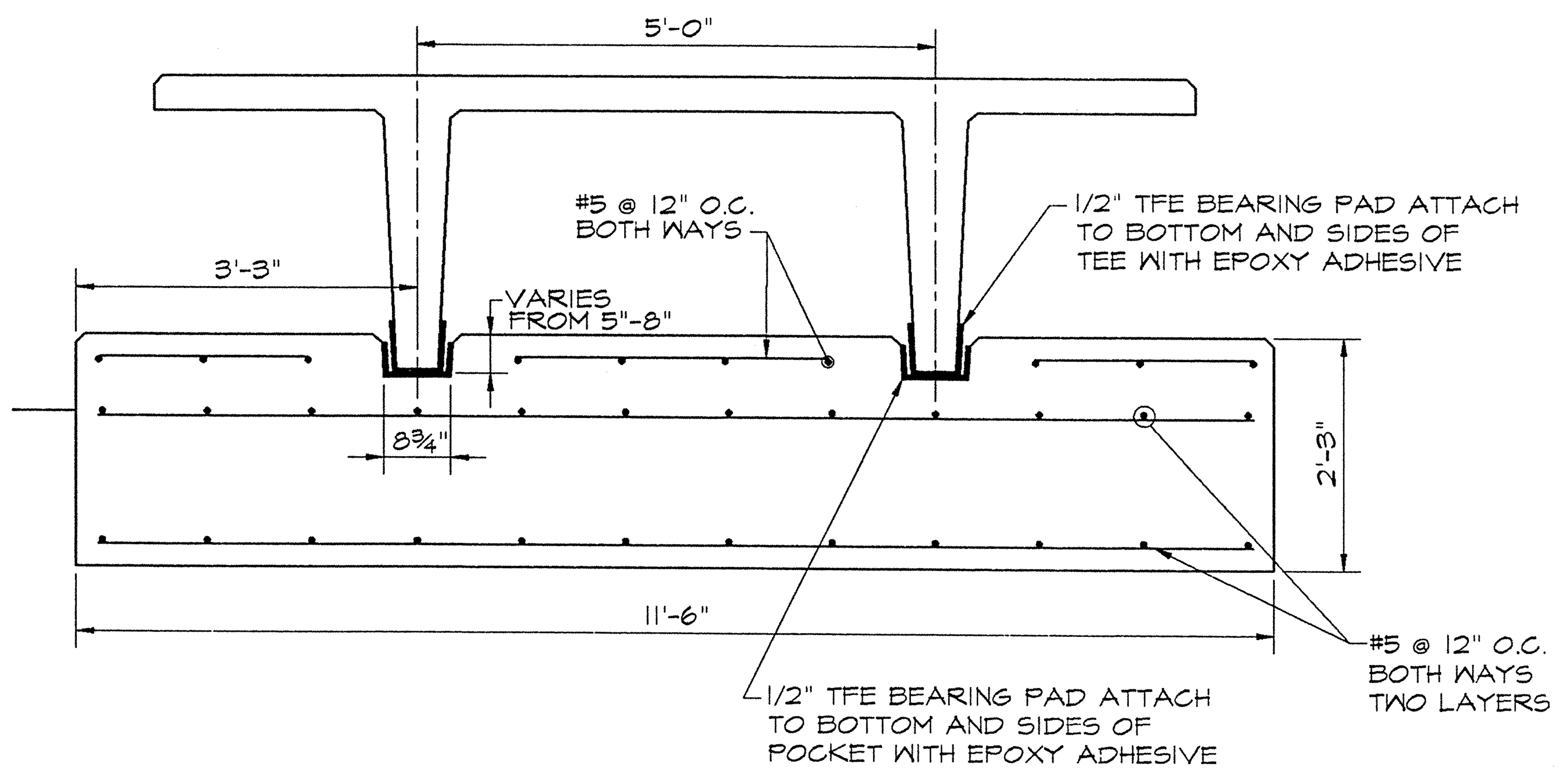
SCALE: 3/4" = 1'-0"



SECTION THROUGH RAMP

2 S2 S2

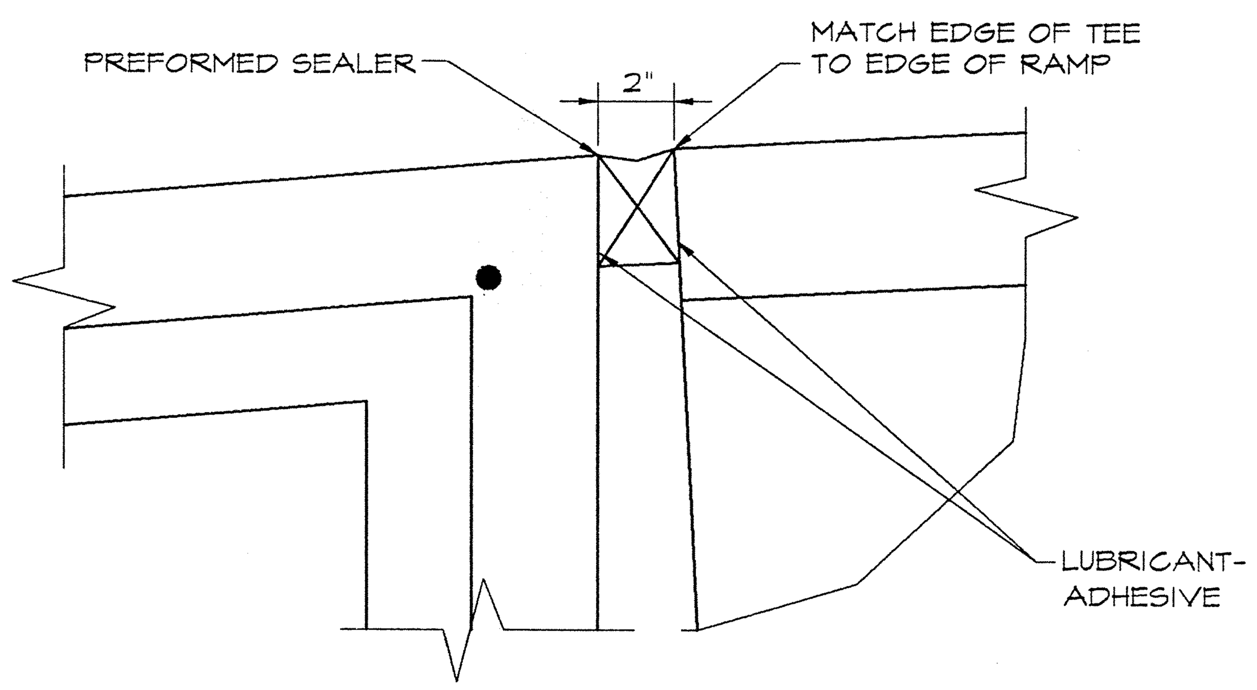
SCALE: 3/4" = 1'-0"



PRECAST DECK FOUNDATION

3 S2 S2

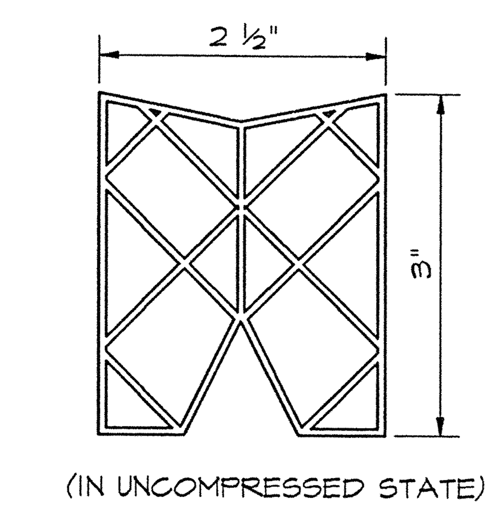
SCALE: 3/4" = 1'-0"



EXPANSION JOINT DETAIL

A S2 S2

SCALE: 3" = 1'-0"

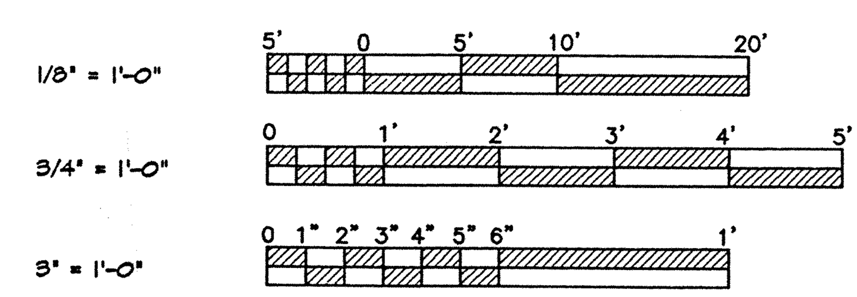


(IN UNCOMPRESSED STATE)

PREFORMED ELASTOMERIC JOINT SEALER

NO TO SCALE

GRAPHIC SCALES



SYM	DESCRIPTION	DATE	APPROVED
△	AS BUILT	7/10/97	[signature]

JOB ORDER NO. 1646544
DES. EJA DR. EJA CHK. MLM
E.I.C. M. MONTGOMERY
BR. MGR. F. B. COLE, JR. P.E.
DIV. DIR R. H. IVES, JR. P.E.
APPROVED: DATE: 6/27/96
FOR C.O. COMMANDER, NAVFAC

SUBMITTED BY:
FIRM MEMBER / PRINCIPAL
APPROVED: SANDRA HAWKINSON DATE: 06/03/96
SATISFACTORY TO ACTIVITY
APPROVED: DATE:
SATISFACTORY TO COMNAVBASE

P W C NORVA

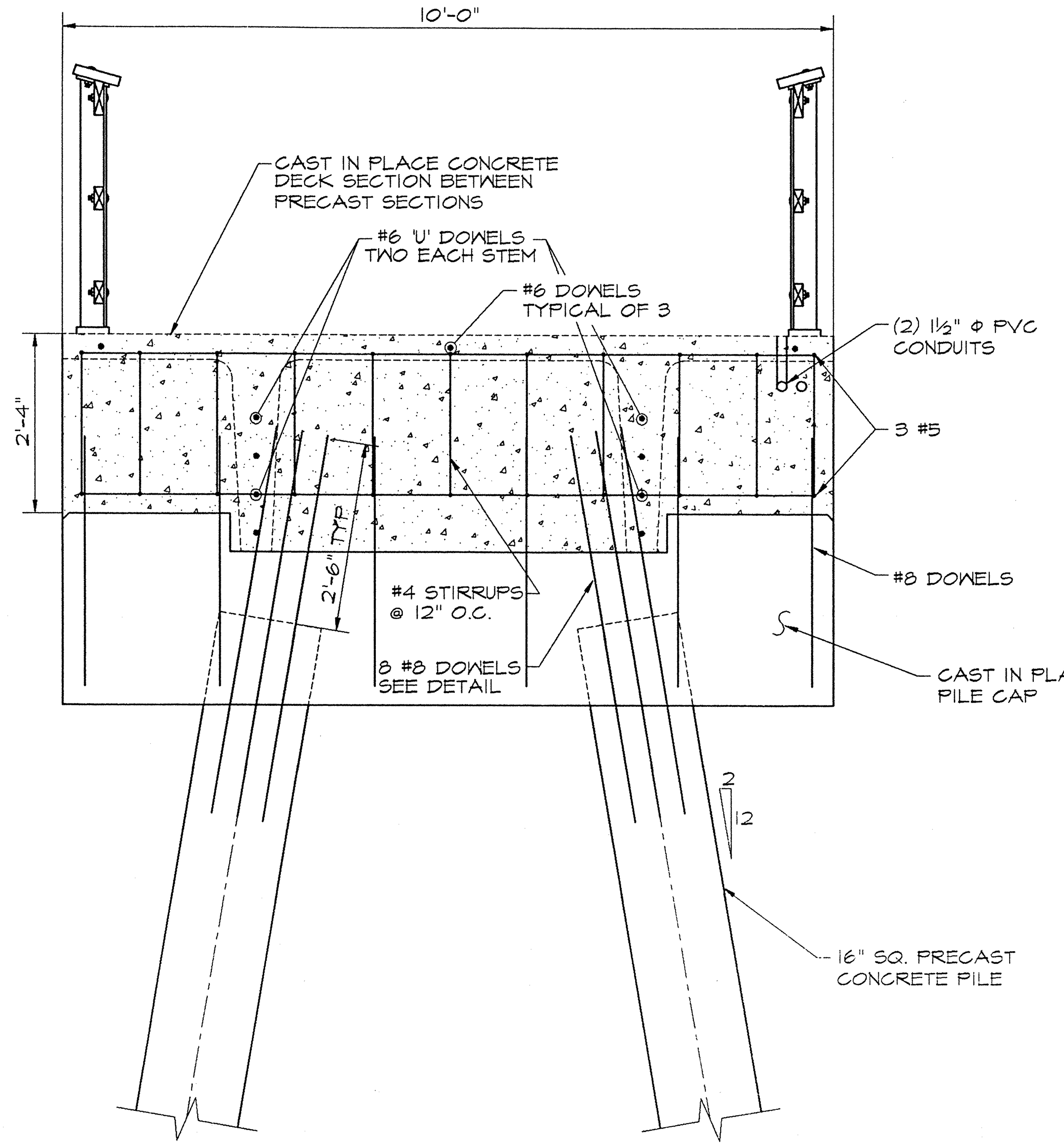
SEAL

DEPT. OF THE NAVY NAVFACENGCOM
NAVY PUBLIC WORKS CENTER
NAVAL STATION NORFOLK, VIRGINIA
NAVALSTATION NORFOLK, VIRGINIA

FISHING PIER
Q-AREA
FOUNDATION PLAN

CODE ID. NO. 80091
DRAWING SIZE: D
CONST. CONT. NO. N62470-96-B-7429
SPEC. 05-96-7429
NAVFAC DRAWING NO. 4,333,339
PWC DRAWING NO. 15,499 D
SHEET 5 OF 13

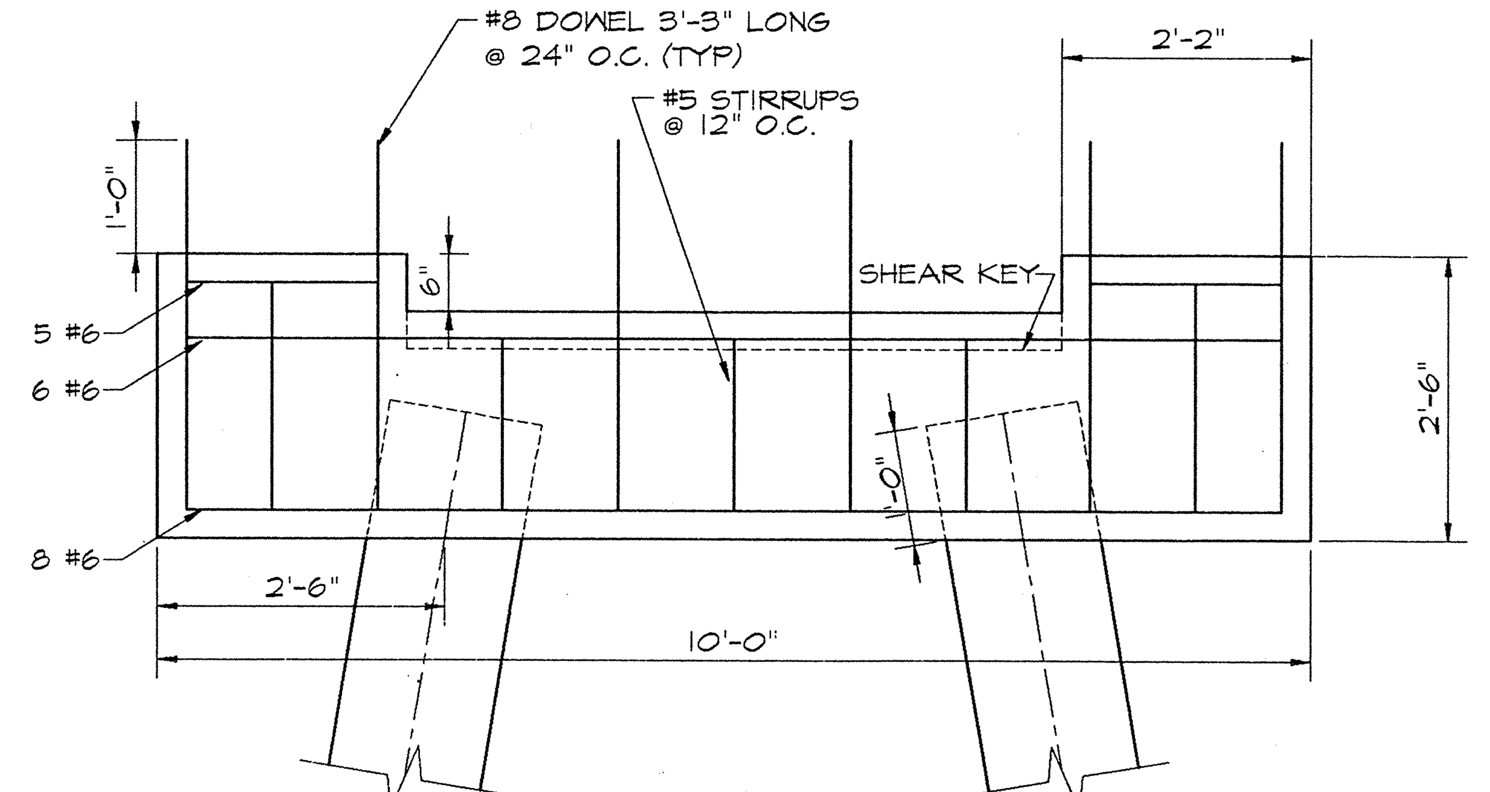
S-2



4
S2 S3

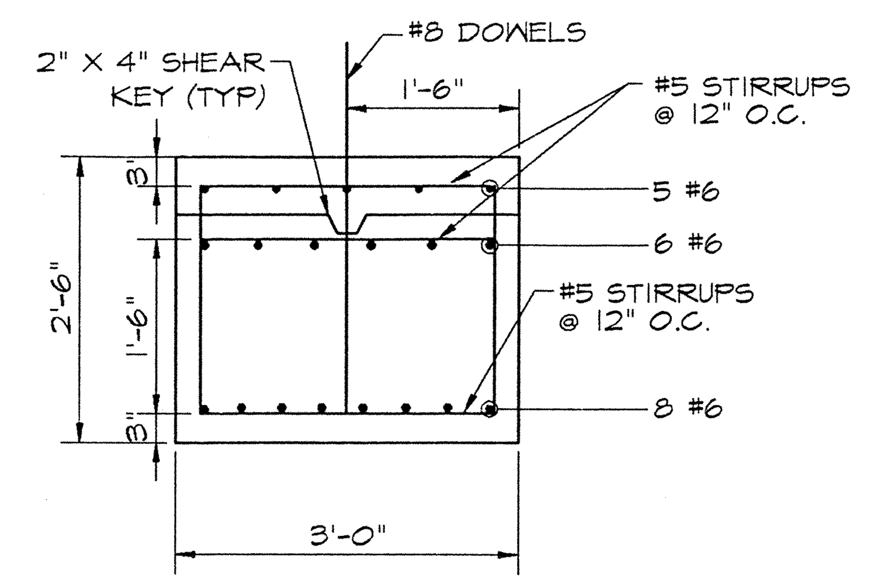
SECTION THROUGH APPROACH

SCALE: 3/4" = 1'-0"



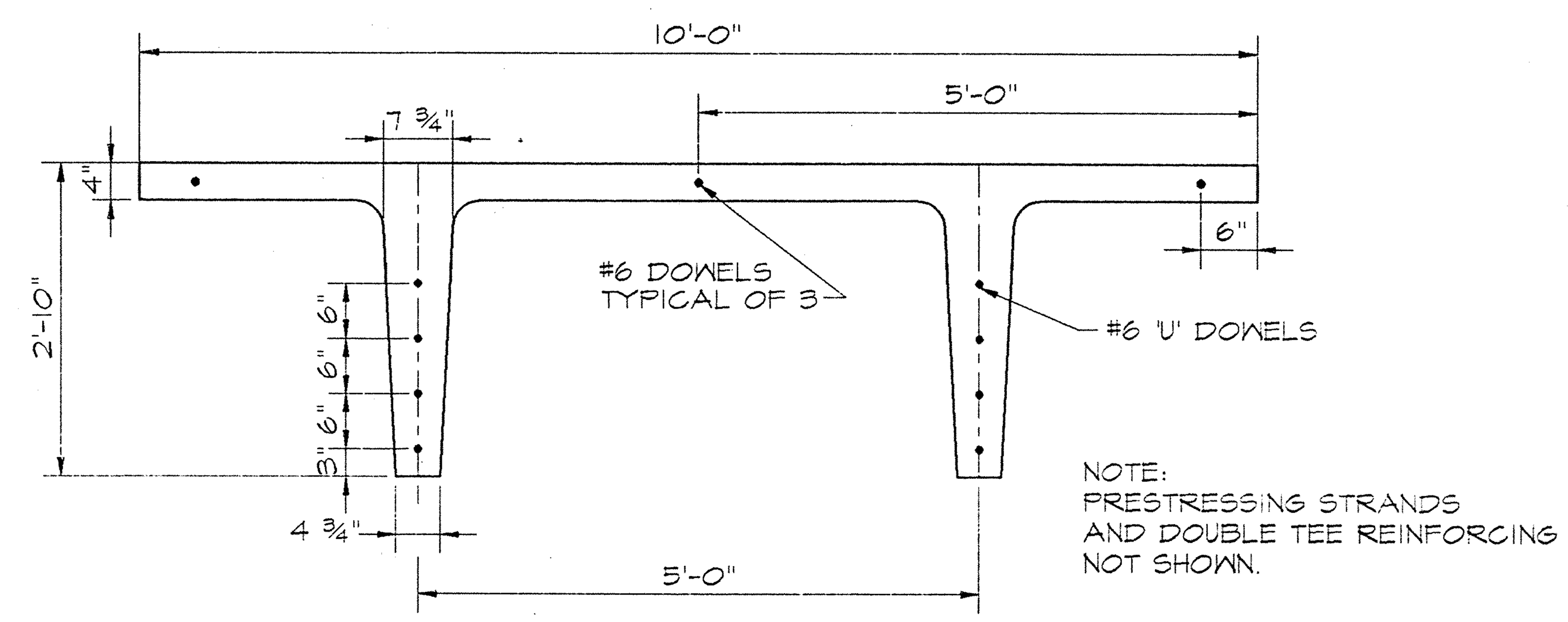
TYPICAL APPROACH PILE CAP

SCALE: 3/4" = 1'-0"



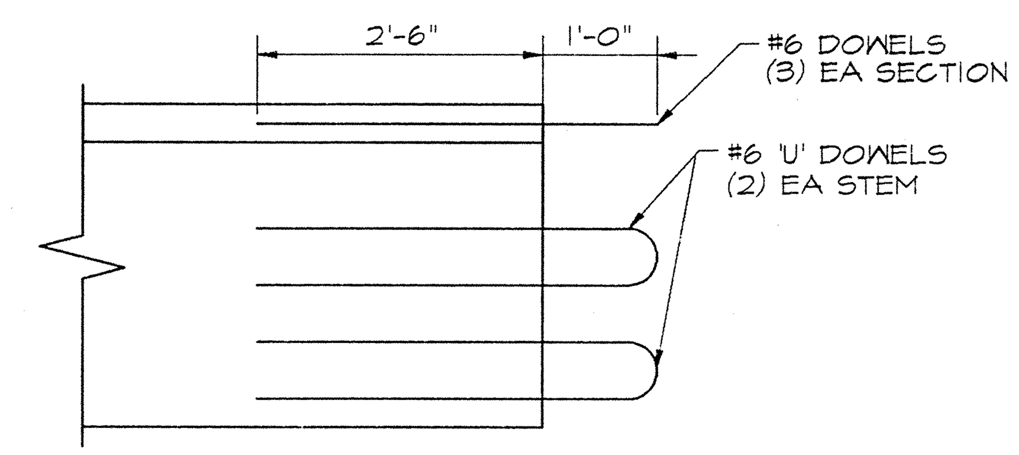
TYPICAL PILE CAP SECTION

SCALE: 3/4" = 1'-0"



TYPICAL PRECAST DOUBLE TEE

SCALE: 3/4" = 1'-0"



TYPICAL PRECAST DOUBLE TEE

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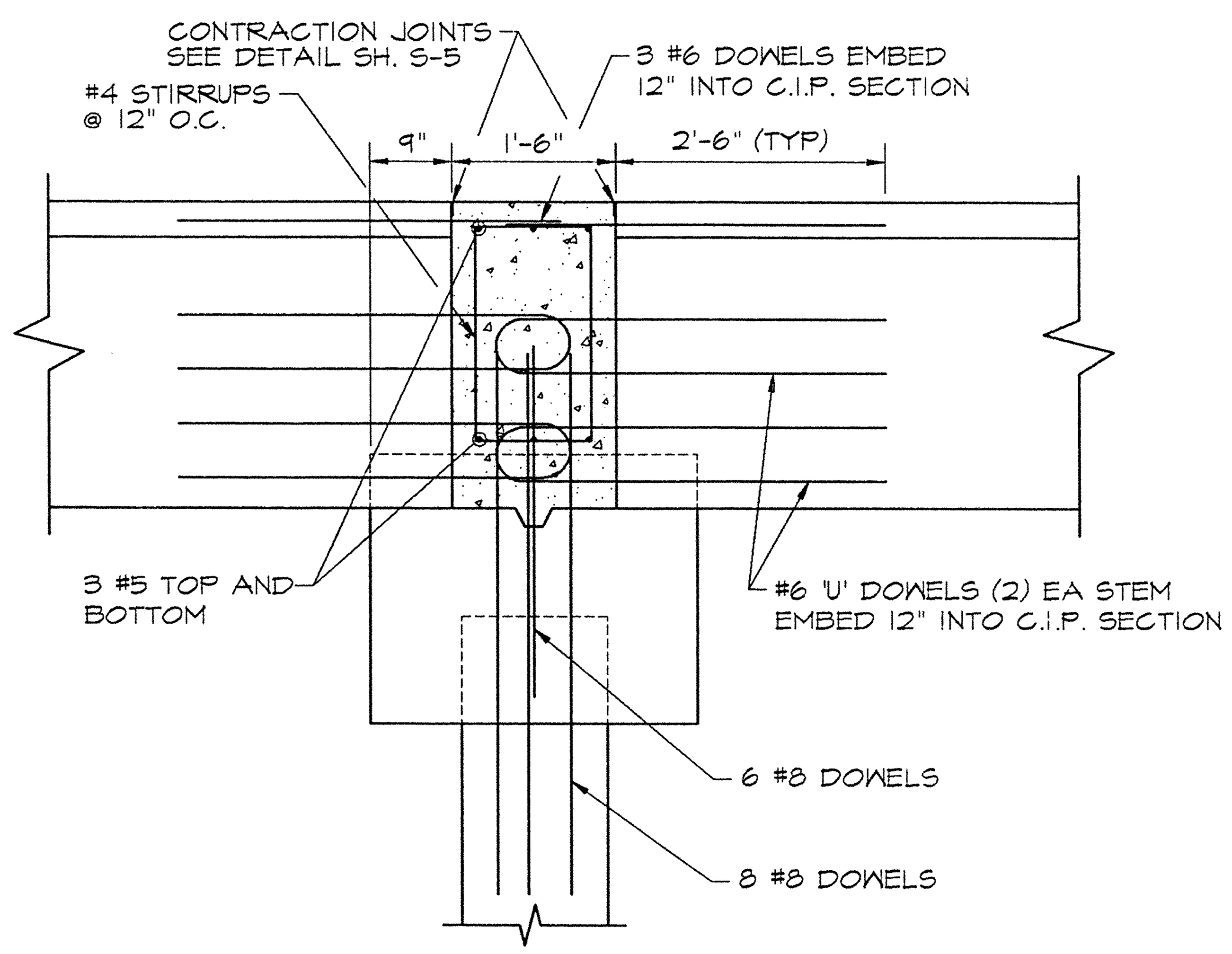
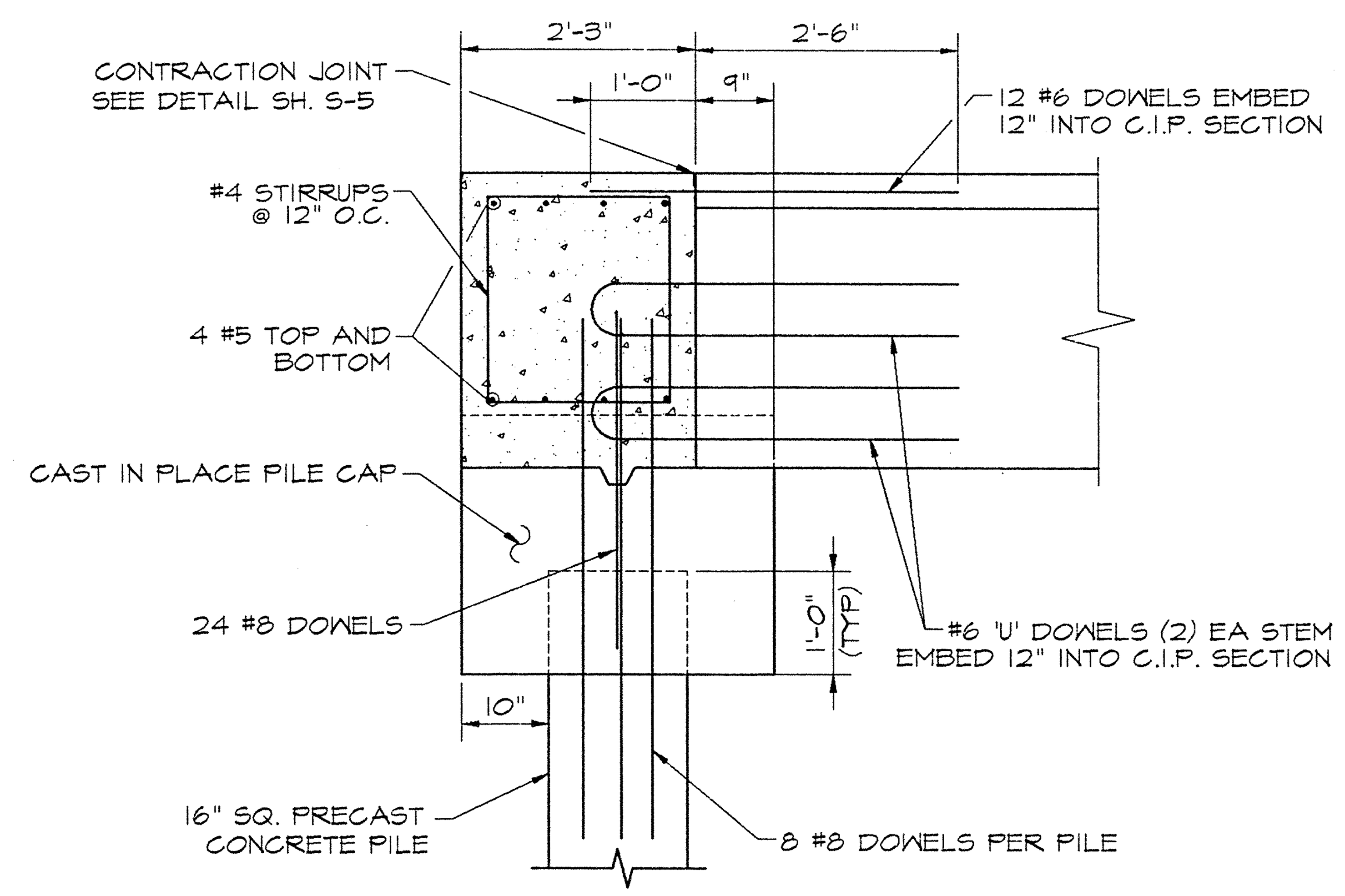


5
S2 S3

CAST-IN-PLACE SECTION

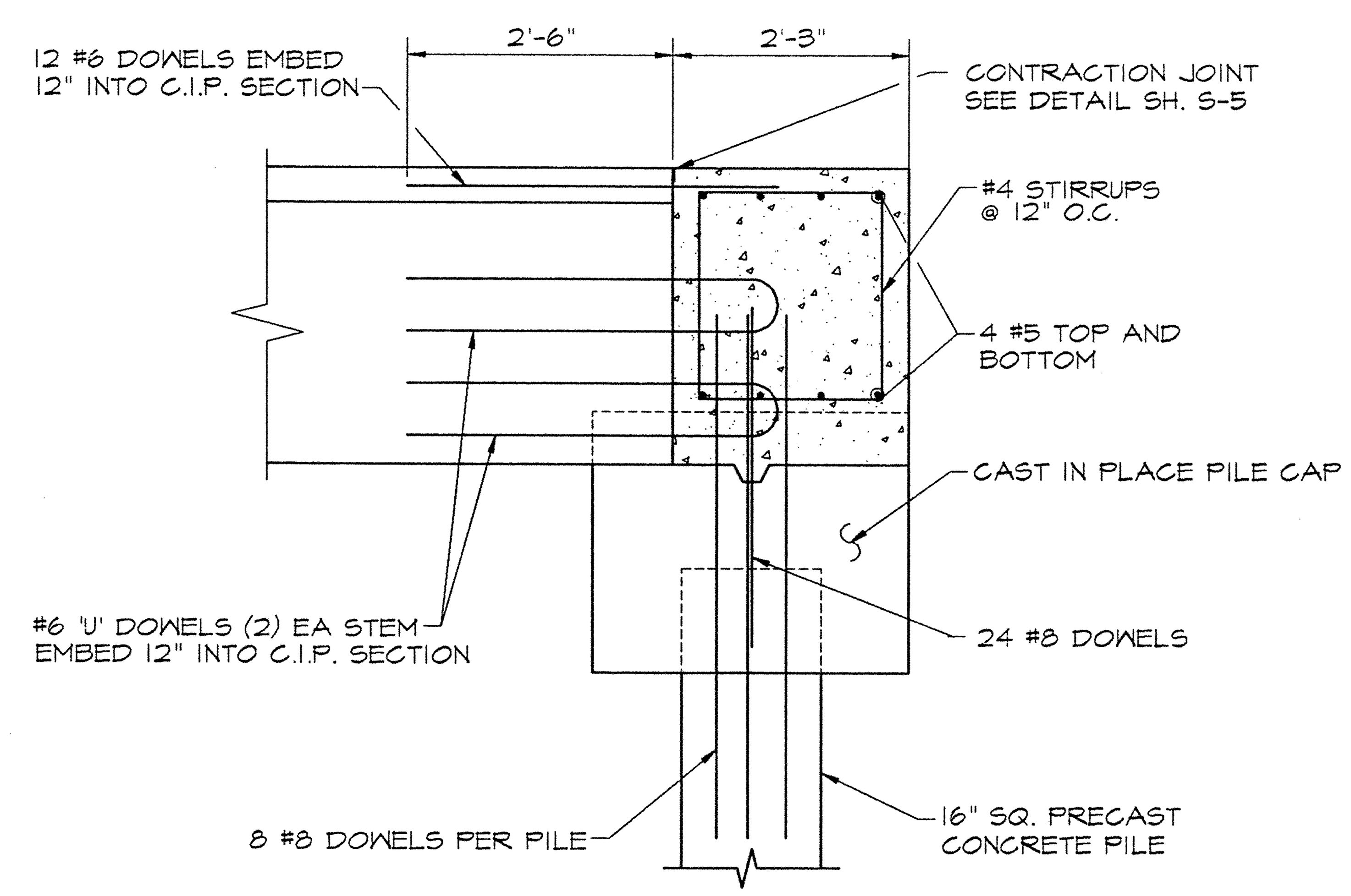
SCALE: 3/4" = 1'-0"



6
S2 S3

END OF PIER SECTION

SCALE: 3/4" = 1'-0"



7
S2 S3

END OF PIER SECTION

SCALE: 3/4" = 1'-0"

GRAPHIC SCALE

0 1' 2' 3' 4' 5'

3/4" = 1'-0"

AS BUILT 7/10/97

SYM	DESCRIPTION	DATE	APPROVED

REVISIONS

SUBMITTED BY: FIRM MEMBER / PRINCIPAL

APPROVED: DATE: SANDRA HANKINSON 08/09/96

SATISFACTORY TO ACTIVITY

APPROVED: DATE:

SATISFACTORY TO COMNAVBASE

JOB ORDER NO. 1646544

DES. EJA DR. EJA CHK. MKM

E.I.C. M. MONTGOMERY

BR. MGR. F. B. COLE, JR. P.E.

DIV. DIR. R. H. IVES, JR. P.E.

APPROVED: DATE: 6/27/96

FOR C.O. COMMANDER, NAVFAC

P W C

SEAL NORVA

DEPT. OF THE NAVY NAVFACENGCOM

NAVY PUBLIC WORKS CENTER

NAVAL STATION NORFOLK, VIRGINIA

NAVAL STATION NORFOLK, VIRGINIA

FISHING PIER Q-AREA

SECTIONS AND DETAILS

CODE ID. NO. 80091

DRAWING SIZE: D

CONST. CONT. NO. N62470-96-B-7429

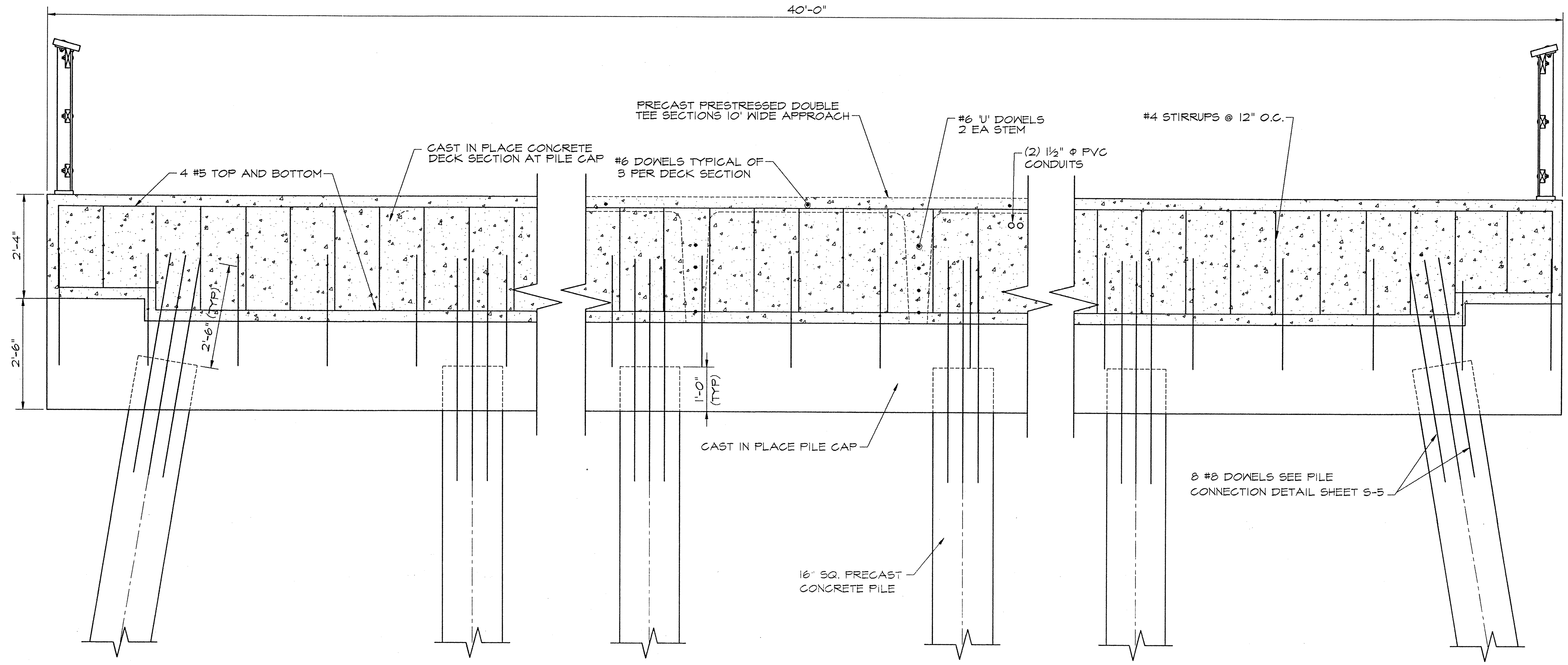
SPEC. 05-96-7429

NAVFAC DRAWING NO. 4333340

PWC DRAWING NO. 15,499 E

SHEET 6 OF 13

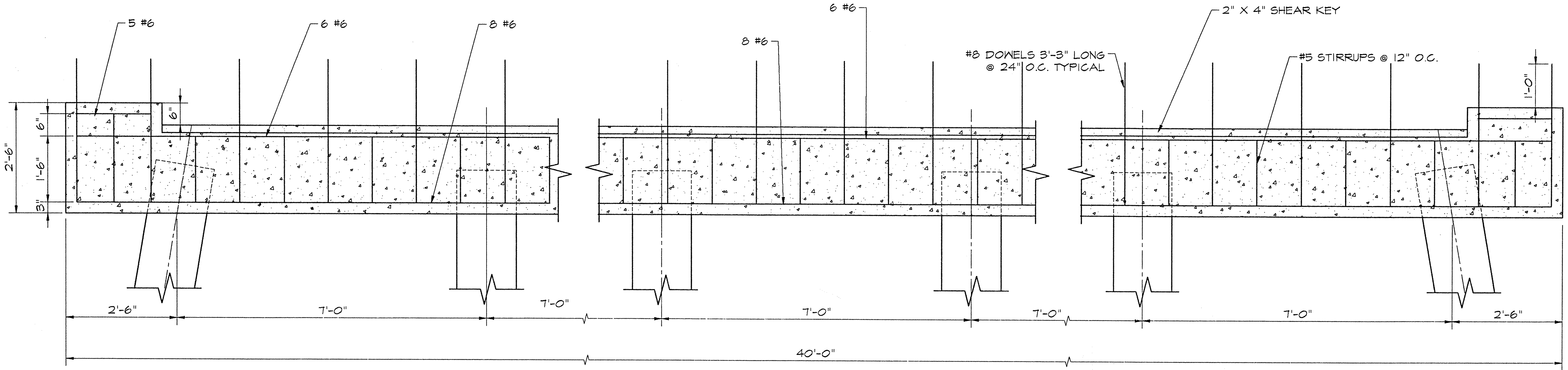
S-3



8 / S2 S4

SECTION THROUGH END OF PIER

SCALE: 3/4" = 1'-0"



END OF PIER PILE CAP

SCALE: 3/4" = 1'-0"

GRAPHIC SCALE

0 1' 2' 3' 4' 5'

3/4" = 1'-0"

SYM	DESCRIPTION	DATE	APPROVED
	AS-BUILT	7/10/97	

SUBMITTED BY:
FIRM MEMBER / PRINCIPAL
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SANDRA HAWKINSON 06/03/96
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PWC NORVA

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NAVY PUBLIC WORKS CENTER
NAVAL STATION NORFOLK, VIRGINIA
NAVAL STATION NORFOLK, VIRGINIA

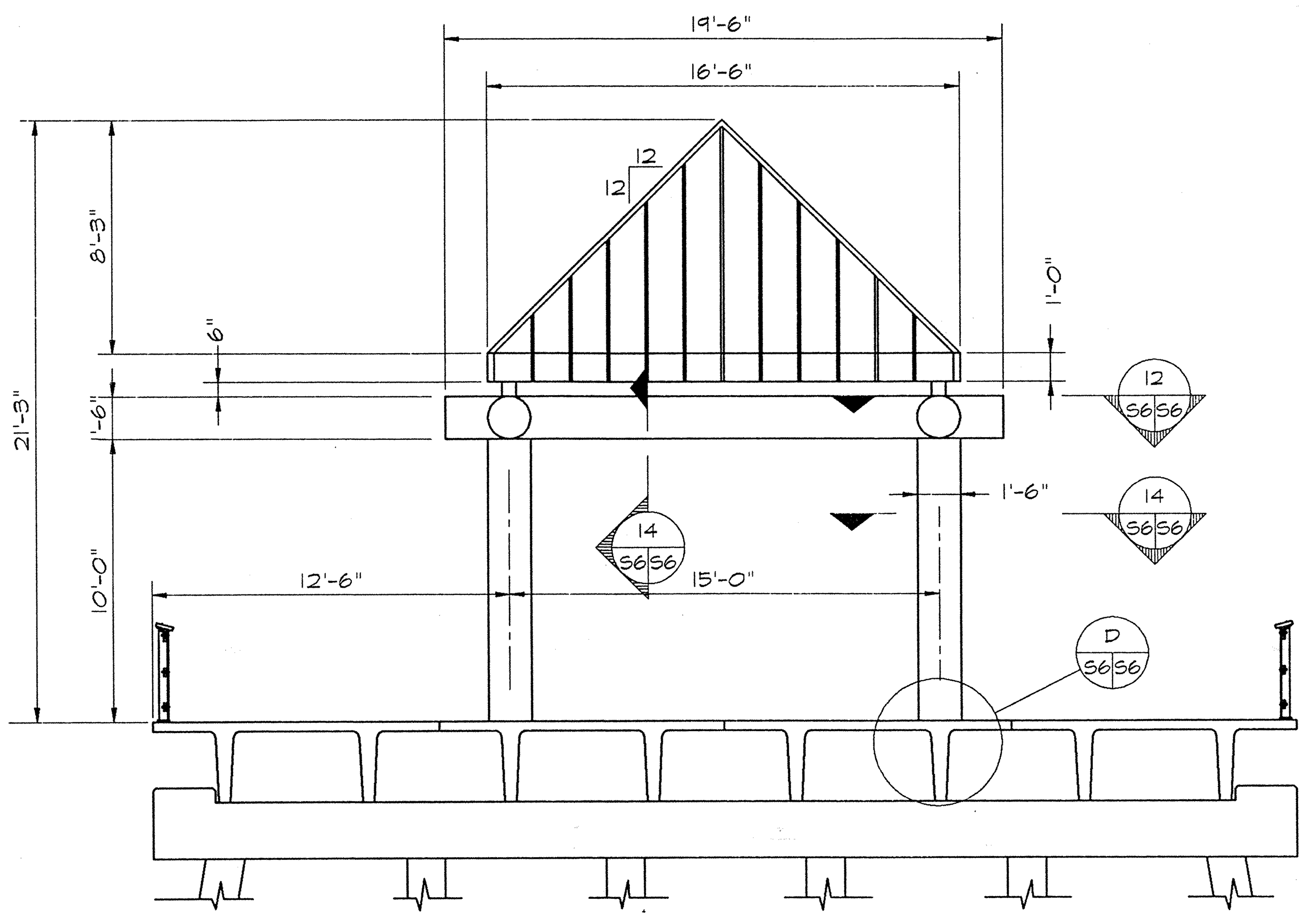
FISHING PIER
Q-AREA

SECTIONS AND DETAILS

CODE ID. NO. 80091
DRAWING SIZE: D
CONST. CONT. NO. N62470-96-B-7429
SPEC. 05-96-7429
NAVFAC DRAWING NO. 4,333,341
PWC DRAWING NO. 15,499 F
SHEET 7 OF 13

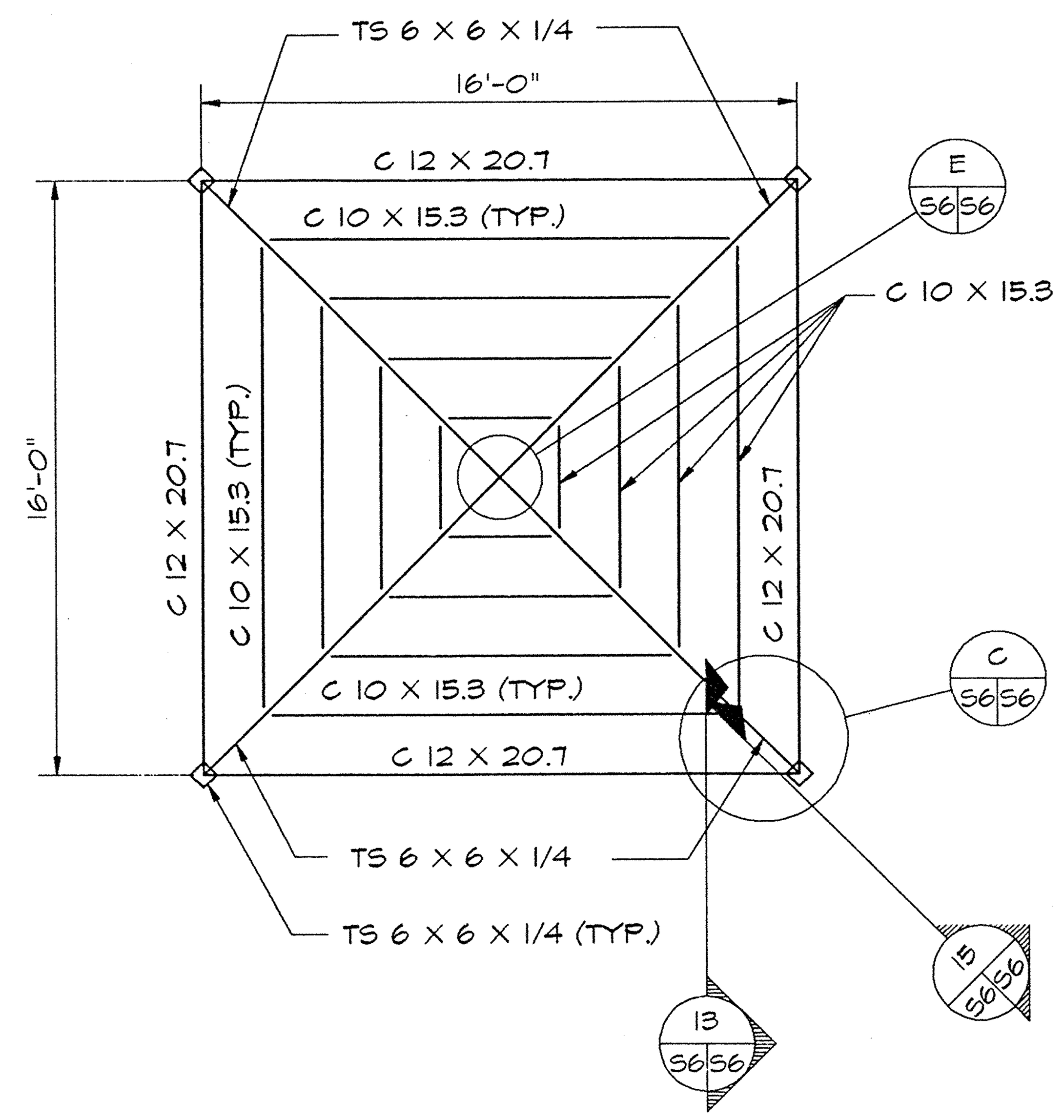
S-4

HANDRAIL POST PLAN
SCALE: 1/8" = 1'-0"
5 SPACES @ 7'-2"
8 SPACES @ 7'-5"
23 SPACES @ 7'-10"
LIGHT BASE TYPICAL
ANGLE POST TYPICAL OF 6 CORNERS
REDUCED HEIGHT OF HANDRAIL TYPICAL OF FOUR LOCATIONS
2 SPACES @ 7'-2"
2 SPACES @ 7'-6"
2 SPACES @ 7'-10"
3'-0" 3'-0"
4'-0" (TYP)
2 SPACES @ 7'-7" (TYP)
3'-10" (TYP)
TYPICAL ANGLE BASE PLATE
SCALE: 3" = 1'-0"
1" ⌀ HOLES FOR ¾" ⌀ BOLTS
L 6 X 6 X 3/8 TYPICAL FOR ALL CORNERS
PL ¾" X 7" X 0'-7"
EDGE OF PIER
5 1/4" (TYP)
TYP
HANDRAIL ELEVATION
PL ½" X 5" X 0'-7" @ 10° ANGLE
WT4 X 12 @ 8'-0" O.C. MAX
PL ½" X 5" X 0'-7"
HANDRAIL SECTION
10d GALVANIZED NAIL 12" O.C.
2X8 TOP RAIL @ 10° ANGLE
2X6 NOTCH TO FIT AROUND TOP PLATE
2X4 RAILS TYP
⅝" ⌀ GALVANIZED CARRIAGE BOLTS IN ½" ⌀ HOLES TYP FOR ALL TIMBER CONNECTIONS
WT4 X 12
⅝" ⌀ GALVANIZED THROUGH BOLT WITH LOCKING NUT AND WASHER
SEAL AROUND PL WITH SILICONE
3'-6" OR 2'-6"
1'-0" (TYP)
EYEBOLTS AND CHAIN W/ LATCHES TYPICAL 4 LOCATIONS
L 6 X 6 X ¼" @ 10° ANGLE
2X8 TOP RAIL @ 10° ANGLE
2X6 NOTCH TO FIT AROUND TOP PLATE
WT4 X 12 @ 8'-0" O.C. MAX
PL ½" X 5" X 0'-7"
3'-6"
2'-6"
1'-6" ⌀
1" CHAMFER
4 #8
4 1/2" (TYP)
#3 STIRRUPS @ 12 O.C.
4 #8 BEND 90° EXTEND INTO FOOTING 12"
3'-0"
AS BUILT
REVISIONS
DESCRIPTION
DATE
APPROVED
SYM
JOB ORDER NO. 1646544
DES. EJA DR. EJA CHK. MLM
E.I.C. M. MONTGOMERY
BR. MGR. F. B. COLE, JR. P.E.
DIV. DIR R. H. IVES, JR. P.E.
SUBMITTED BY:
FIRM MEMBER / PRINCIPAL
APPROVED: SANDRA HAWKINSON 6/13/96
SATISFACTORY TO ACTIVITY
FOR C.O. COMMANDER, NAVFAC
SATISFACTORY TO COMNAVBASE
P W C
NORVA
SEAL
NAVFACENGCOM
NORFOLK, VIRGINIA



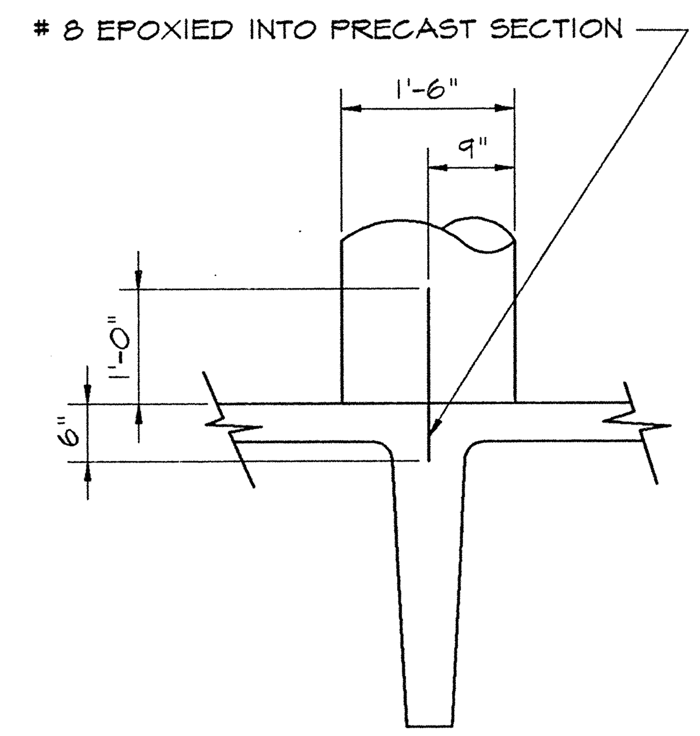
SECTION - GAZEBO

SCALE: NONE



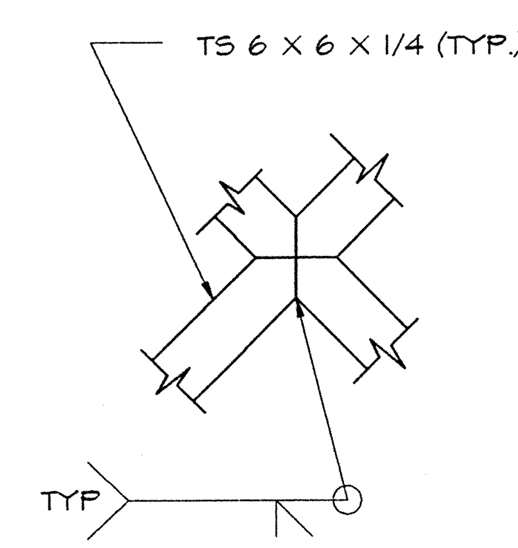
ROOF FRAMING PLAN

SCALE: NONE



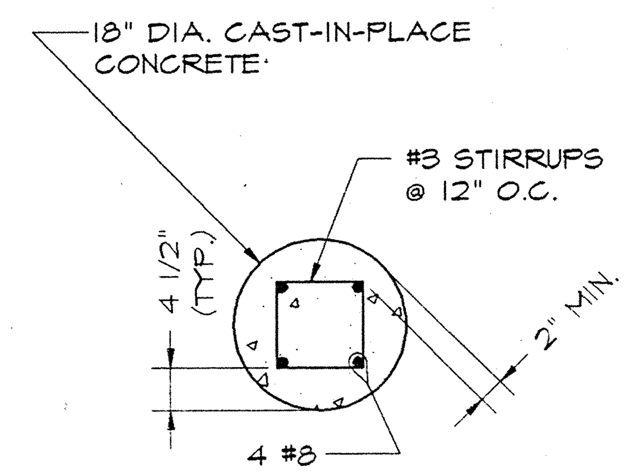
D S6 S6 DETAIL

SCALE: NONE



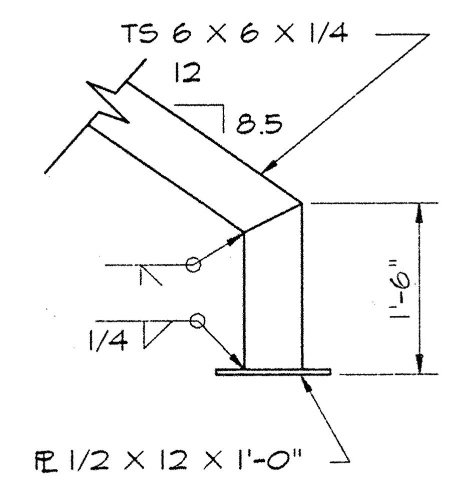
E S6 S6 DETAIL

SCALE: NONE



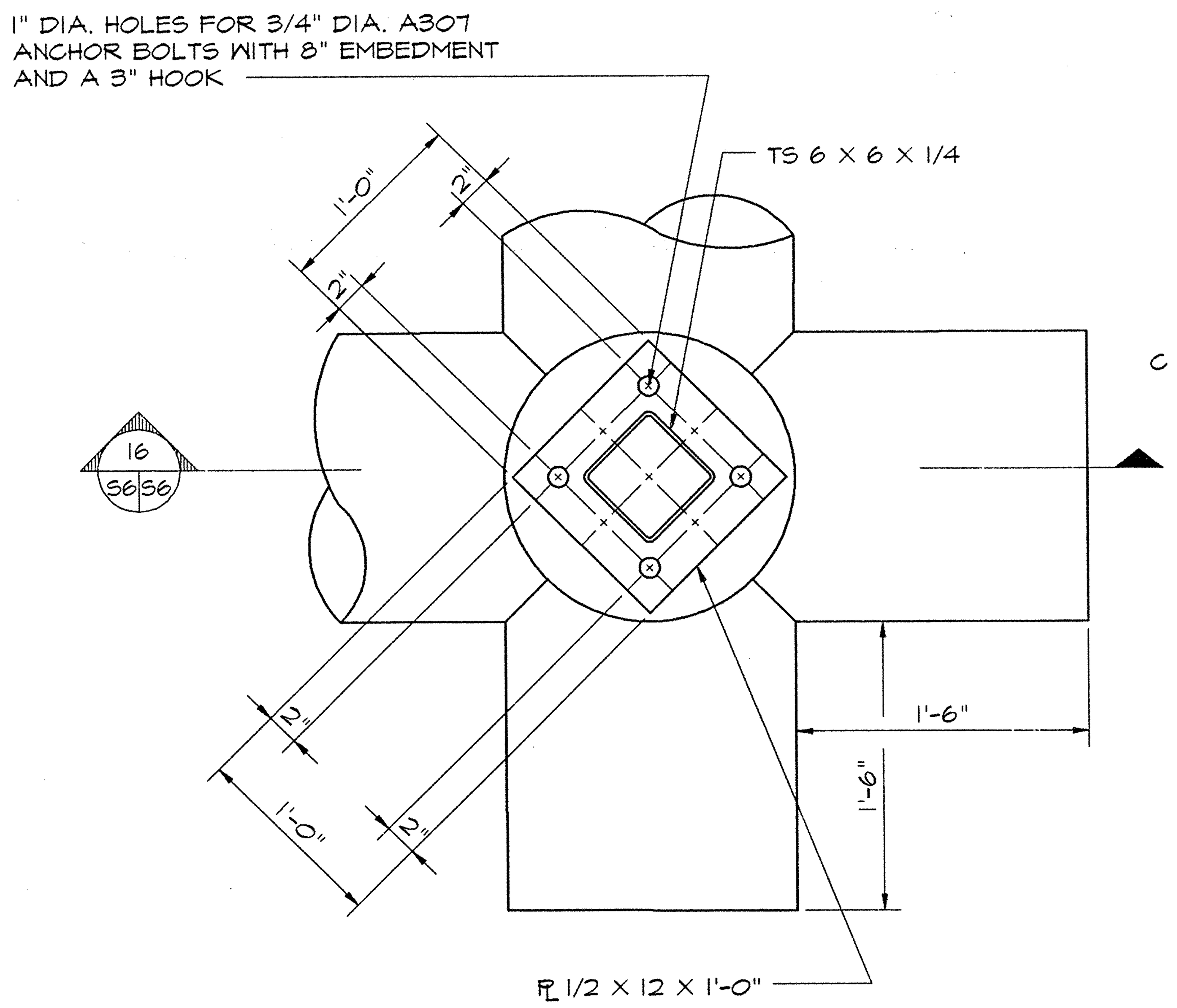
14 S6 S6 SECTION

SCALE: NONE



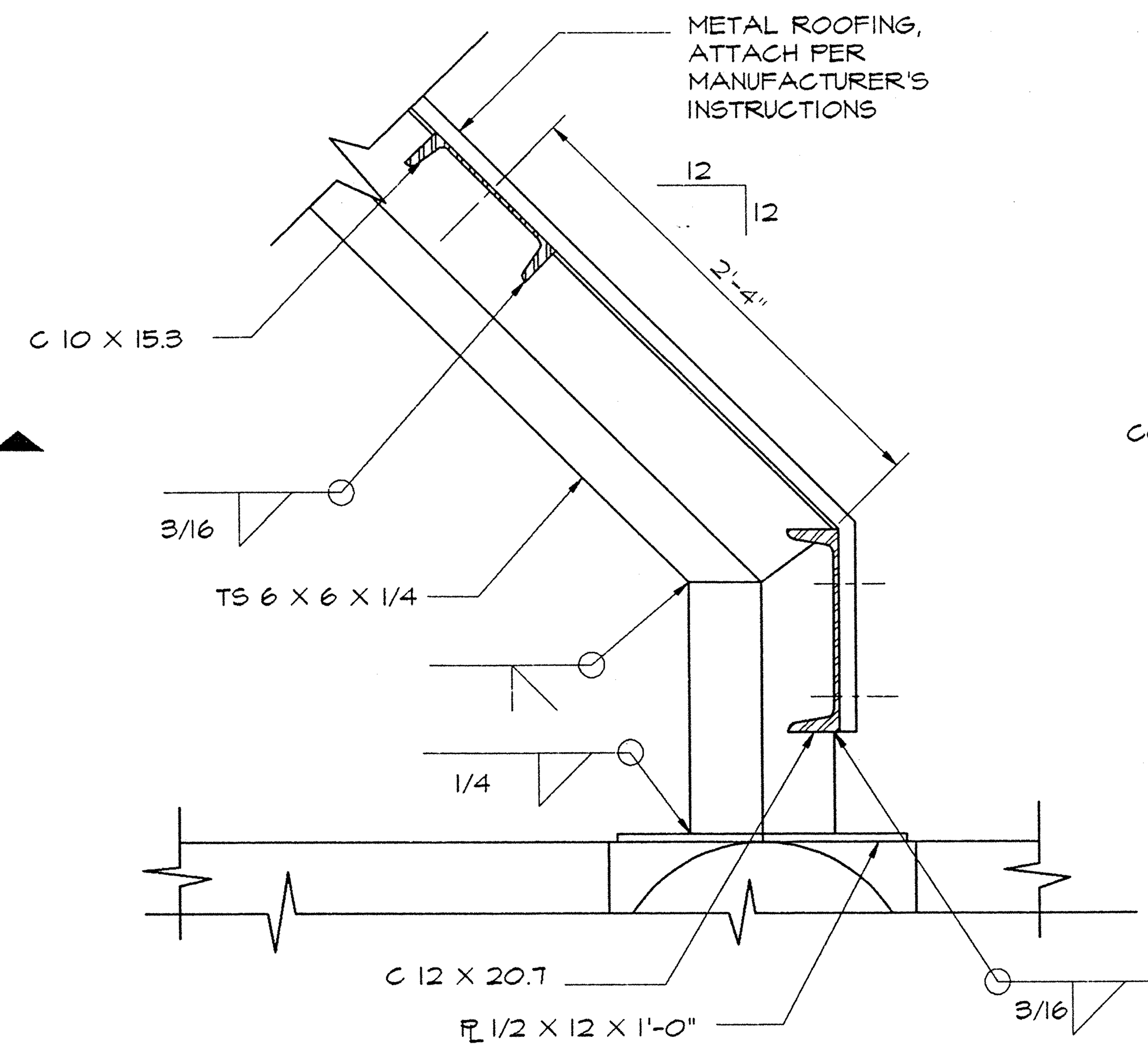
15 S6 S6 SECTION

SCALE: NONE



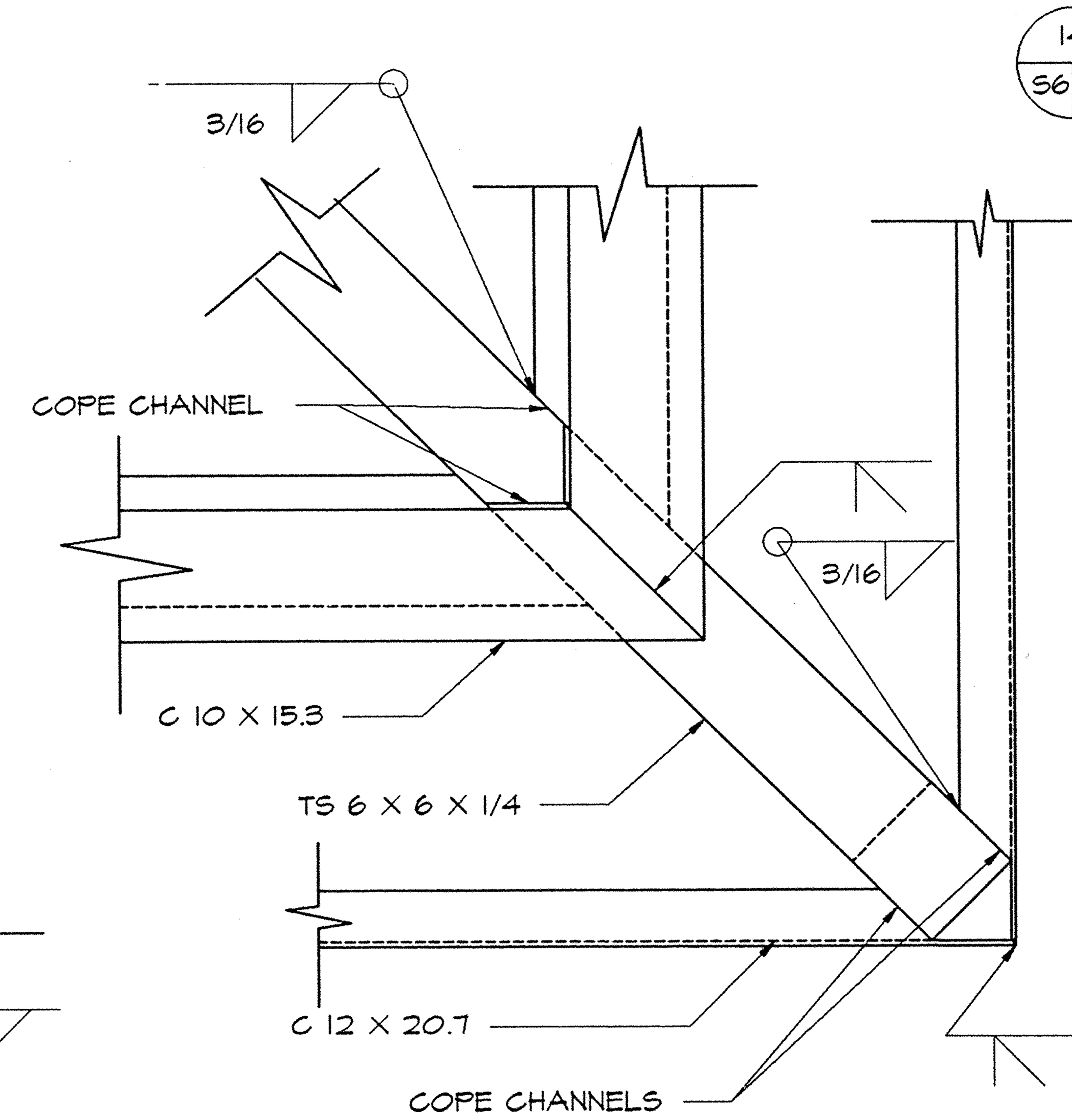
12 S6 S6 SECTION

SCALE: NONE



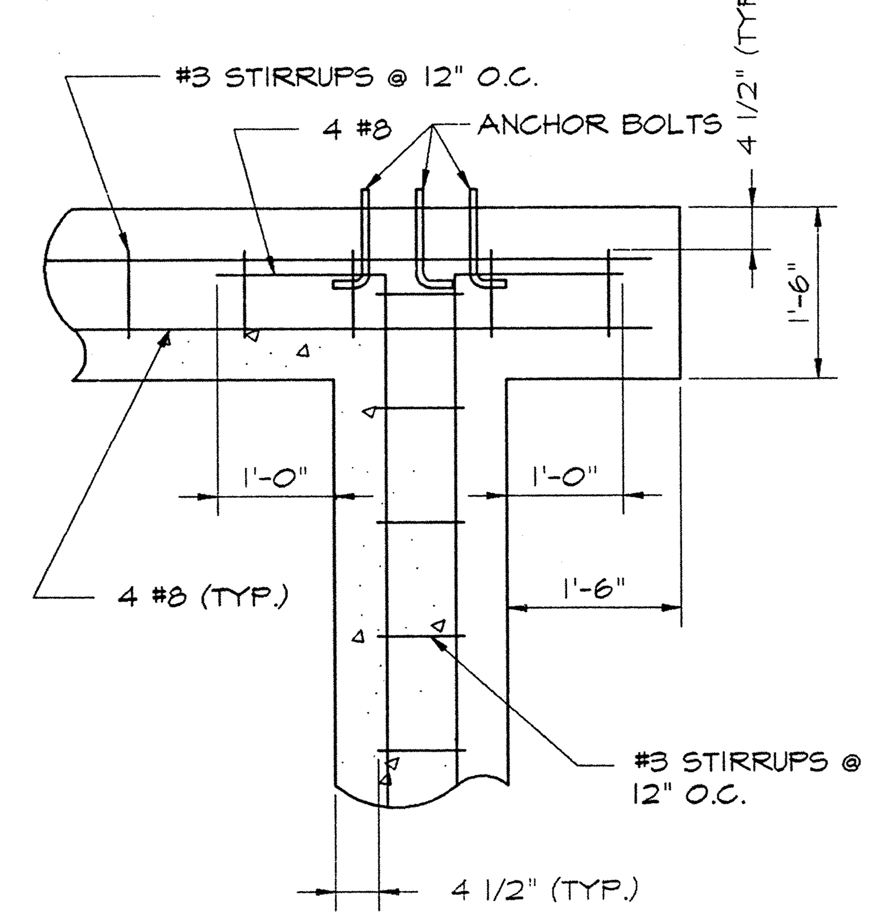
13 S6 S6 SECTION

SCALE: NONE



C S6 S6 SECTION

SCALE: NONE



16 S6 S6 SECTION

SCALE: NONE

AS BUILT 7/10/97

SYM	DESCRIPTION	DATE	APPROVED

REVISIONS

SUBMITTED BY:
FIRM MEMBER / PRINCIPAL
APPROVED: SANDRA HAWKINSON DATE: 6/3/96
SATISFACTORY TO ACTIVITY
APPROVED: DATE:
SATISFACTORY TO COMNAVBASE

JOB ORDER NO. 1646544
DES. MKM DR. MKM CHK. EJA
E.I.C. M. MONTGOMERY
BR. MGR. F. B. COLE, JR. P.E.
DIV. DIR. R. H. IVES, JR. P.E.
APPROVED: DATE:
FOR CO, COMMANDER, NAVFAC

PWC NORVA

SEAL

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NAVY PUBLIC WORKS CENTER
NAVAL STATION NORFOLK, VIRGINIA
NAVAL STATION NORFOLK, VIRGINIA
FISHING PIER Q-AREA
GAZEBO SECTIONS AND DETAILS

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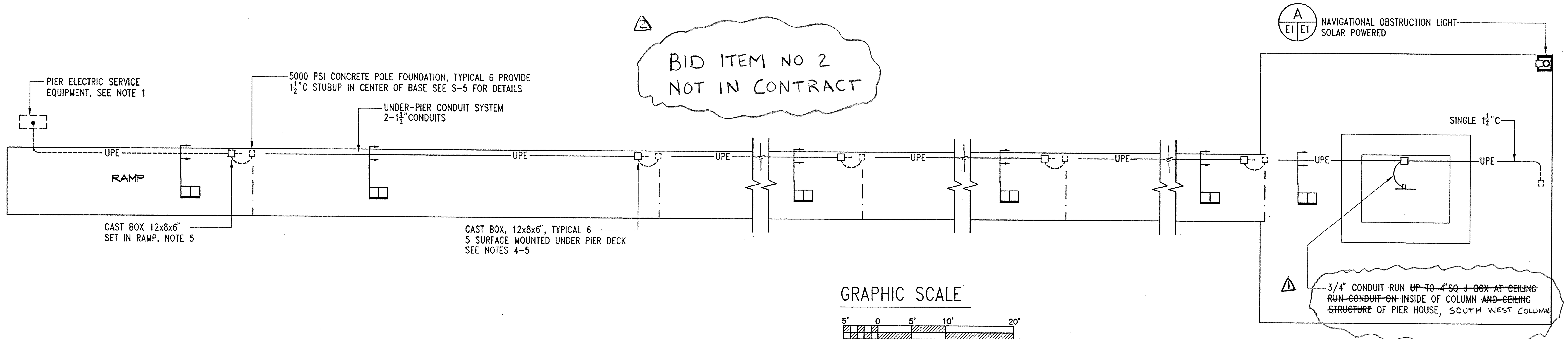
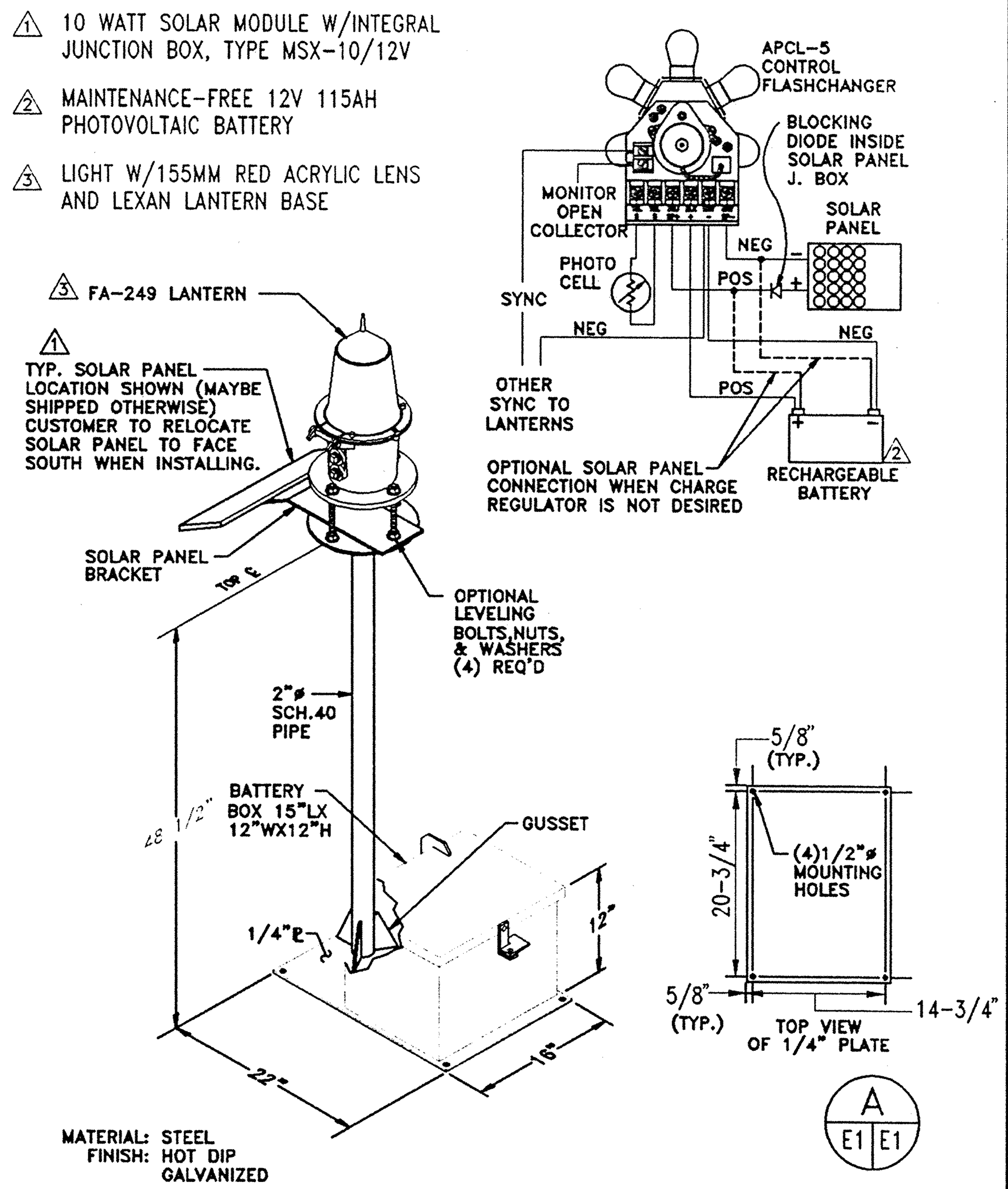
DRAWING SIZE: D
CONST. CONT. NO. N62470-96-B-7429
SPEC. 05-96-7429
NAVFAC DRAWING NO. 4,333,343
PWC DRAWING NO. 15,499 H
SHEET 9 OF 13

S-6

PIER - ELECTRICAL PLAN

SCALE: 1/8" = 1'-0"





NAVIGATIONAL OBSTRUCTION LIGHT

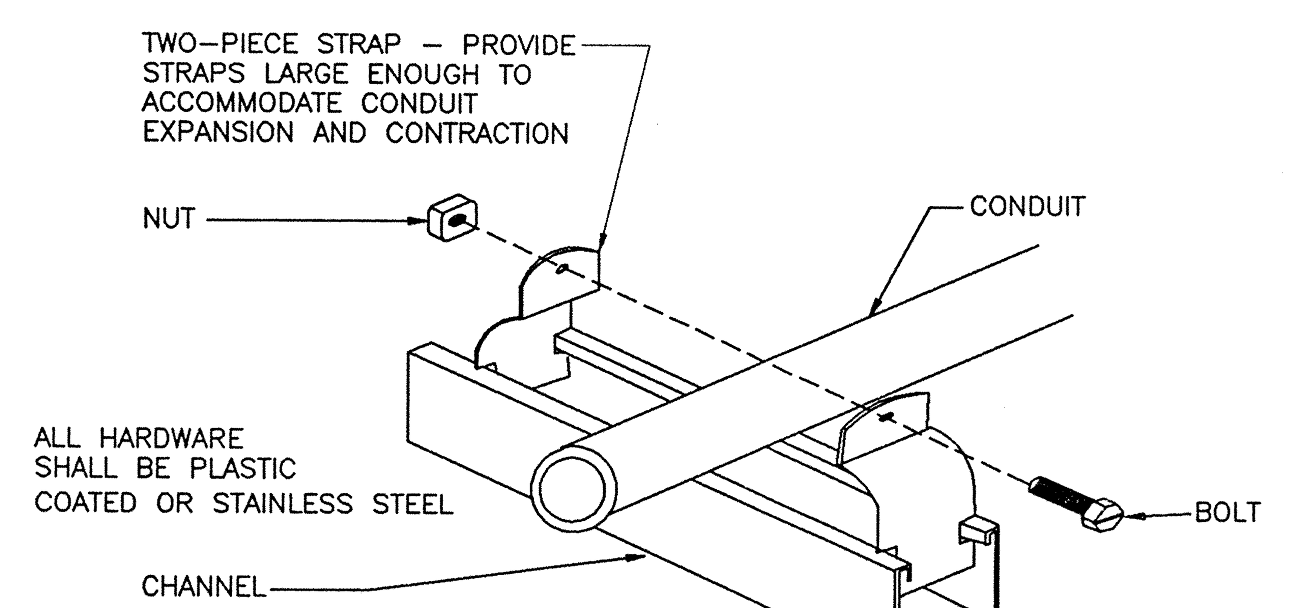
SKETCH DATE	JUNE 1996 /kjf	STYLE	XL-2D

LEGEND

———UPE———	UNDER PIER ELECTRIC CONDUIT
- - - - - - -	CONDUIT CAST IN CONCRETE
———UE———	UNDERGROUND ELECTRIC CONDUIT DASHED INDICATES EXISTING
	DUCTBANK SECTION LOOKING IN DIRECTION OF ARROWS
1 | 2 | 3	CABLE DESIGNATION (TYPICAL) - SEE CABLE SCHEDULE, BLANK INDICATES SPARE HEAVY LINE INDICATES BOTTOM OF DUCT
	PRECAST CONCRETE HANDHOLE WITH CONCRETE COVER
	JUNCTION BOX, SIZE INDICATED
	EXISTING PAD MOUNT TRANSFORMER 112.5KVA 11.5KV - 208Y/120V
	SERVICE PANELBOARD, 120/208V 3P-4W 100A MCB NEMA 3X ENCLOSURE, W/UL SERVICE LABEL
	POLE W/2 TYPE A LIGHT FIXTURES 400W METAL HALIDE
	PENDANT MOUNT TYPE B FIXTURE 175W METAL HALIDE
A / E1 | E4	DETAIL DESIGNATION; SHEET WHERE DRAWN; SHEET WHERE CUT

NOTES

1. EXTEND 2-1½" CONDUITS FROM PIER TO SERVICE EQUIPMENT LOCATION. RUN CONDUITS 24" BELOW GRADE AND CAP 6" BELOW GRADE. PIER ELECTRIC SERVICE, WIRING, AND POLE LIGHTS ARE PROVIDED UNDER BID ITEM 2 (SHEETS E-2 THRU E-4).
2. INSTALL SCHEDULE 80 RIGID PVC CONDUIT SURFACE MOUNTED UNDER PIER. PROVIDE EXPANSION COUPLINGS EVERY 20' AND SUPPORT TIGHTLY TO UNDERSIDE OF PRECAST DECK WITH HANGER STRAPS.
3. ATTACH CONDUIT UNDER PIER DECK WITH CHANNEL AND STRAPS, MAXIMUM SPACING 5' BETWEEN SUPPORTS AND WITHIN 3' OF BOXES AND PILE CAPS. TIGHTEN STRAPS ONLY ENOUGH TO ALLOW FORLINEAL MOVEMENT CAUSED BY EXPANSION AND CONTRACTION. EACH CHANNEL SHALL BE HUNG WITH TWO STAINLESS STEEL AND THREADED ANCHORS SET IN CONCRETE GIRDERS.
4. PROVIDE CONDUIT SYSTEM UNDER PIER WITH 5 SURFACE MOUNT CAST IRON BOXES PREDRILLED FOR 5 1½" CONDUIT ENTRANCES. BOXES SHALL BE SIZE 12x8x6"DP WITH MIN 1/4" GALVANIZED IRON PLATE, FOUR MOUNTING LUGS, AND SCREW COVER (STEEL CITY S-120806-A). SURFACE MOUNT BOXES ON UNDERSIDE OF PIER DECK AT LEAST 3' FROM PILECAPS.
5. PROVIDE CAST BOX AND CONDUITS IN RAMP. SET BOX TOP FLUSH WITH DECK WITH COVER ACCESSIBLE FROM ABOVE. BOX IS FLANGED TYPE ER-120806A WITH 5 PREDRILLED 1½"C SLIP HOLES.
6. PROVIDE 1½" CONDUITS THROUGH CAST CONCRETE PILE CAPS AND IN RAMP WITH MINIMUM OF 3" CONCRETE COVER.
7. PROVIDE CONCRETE POLE BASE WITH 1½"C STUBBED UP AND CAPPED. LIGHTS AND OUTLETS ARE TO BE PROVIDED AS BID ITEM 2.
8. ALL SURFACE MOUNT CONDUIT ABOVE PIER SHALL BE PVC COATED RIGID GALVANIZED STEEL CONDUIT WITH PVC COATED FITTINGS AND BOXES.



CONDUIT STRAP DETAIL

NO SCALE

SYM	DESCRIPTION	DATE	APPROVED
⚠	AS BUILT	7/10/97	

REVISIONS

JOB ORDER NO. 1646544
DES. KJF | DR. KJF | CHK. AMD
E.I.C. MKM
BR. MGR. D. N. ROGERS
DIV. DIR R. H. IVES, JR
APPROVED DATE
FOR C.O. COMMANDER, NAVFAC

SUBMITTED BY:
FIRM MEMBER / PRINCIPAL
APPROVED: SANDRA HAWKINSON DATE: 6/3/96
SATISFACTORY TO ACTIVITY
APPROVED: DATE:
SATISFACTORY TO COMNAVBASE

P W C NORVA SEAL

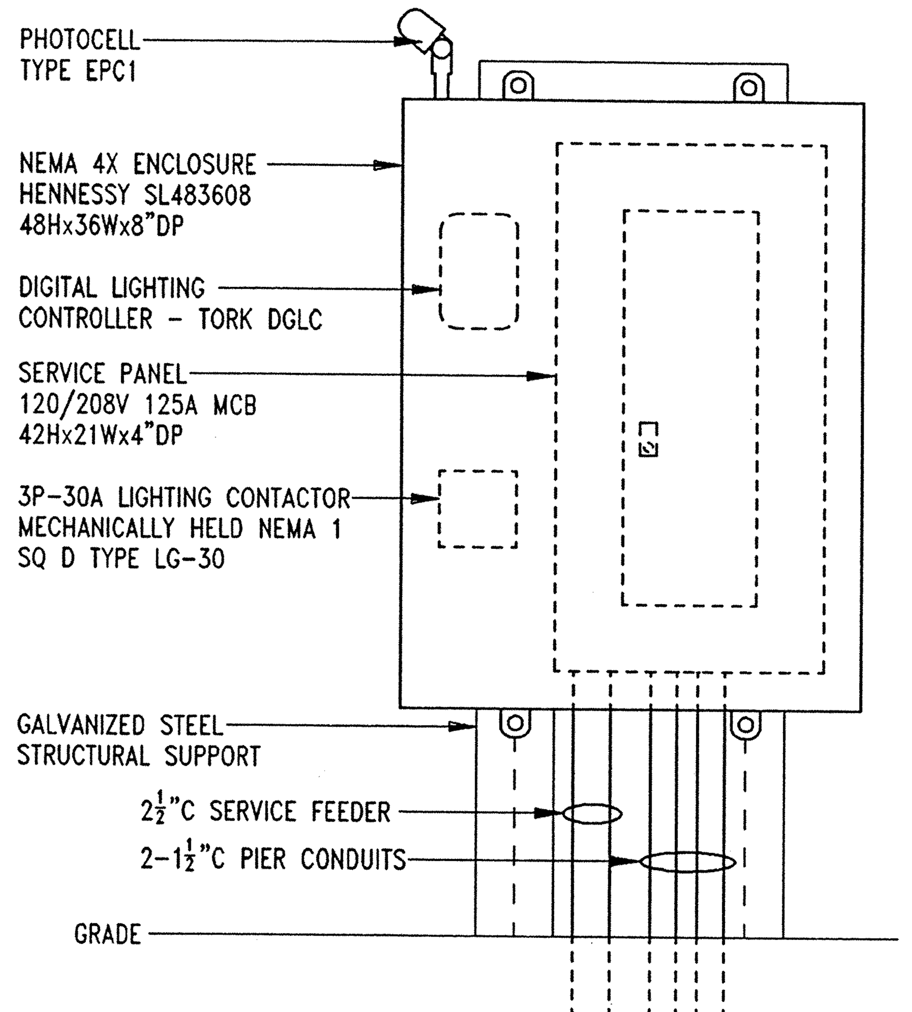
DEPT. OF THE NAVY NAVFACENGCOM
NAVY PUBLIC WORKS CENTER NORFOLK, VIRGINIA
NAVAL STATION NORFOLK, VIRGINIA
NAVAL STATION

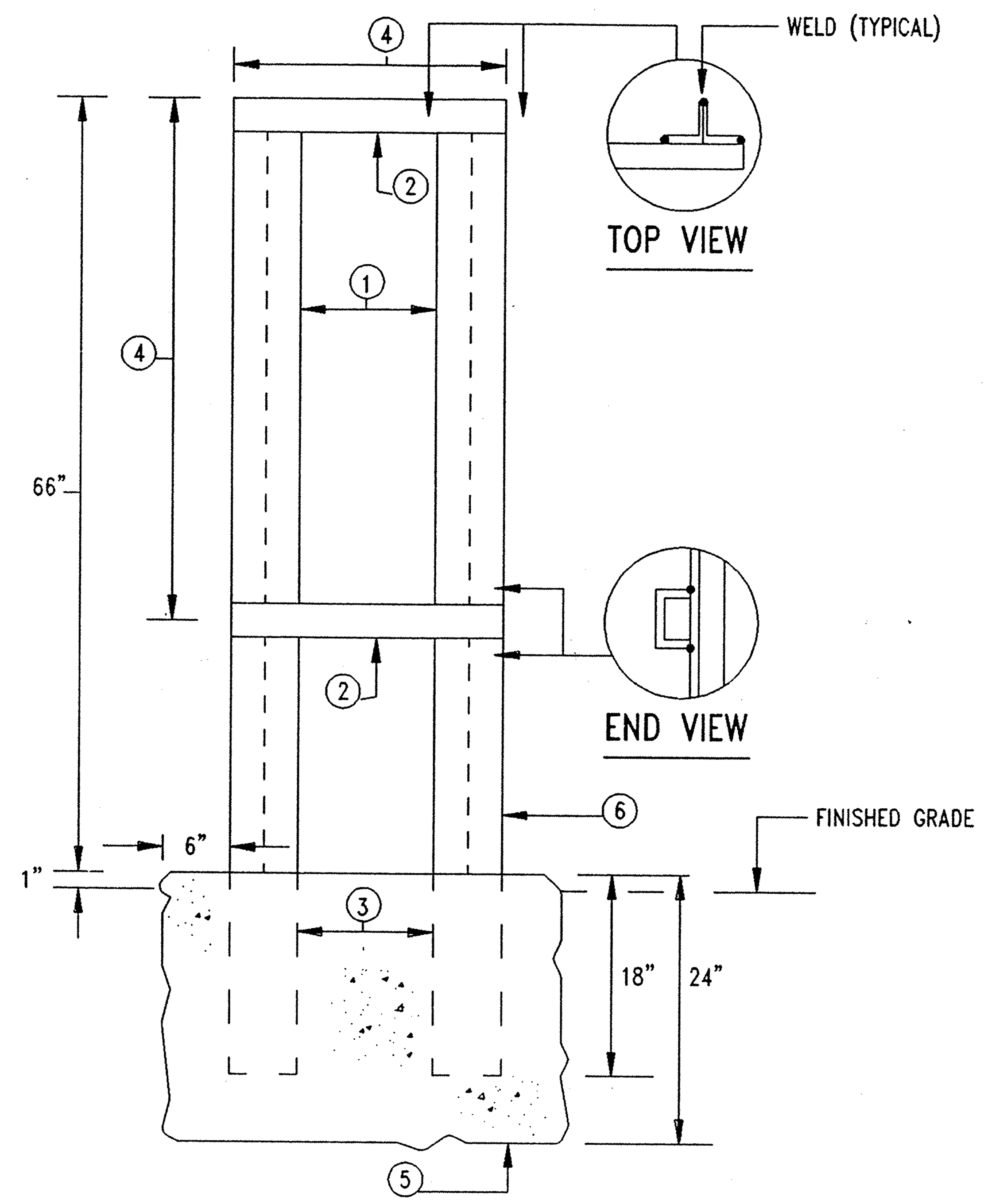
FISHING PIER Q-AREA
PIER ELECTRICAL PLAN

CODE ID. NO. 80091
DRAWING SIZE: D
CONST. CONT. NO. N62470-96-B-7429
SPEC. 05-96-7429
NAVFAC DRAWING NO. 4333344
PWC DRAWING NO. 15,4991
SHEET 10 OF 13

E-1

OUTDOOR ENCLOSURE ON STRUCTURAL SUPPORT
EQUIPMENT INSIDE ENCLOSURE INDICATED



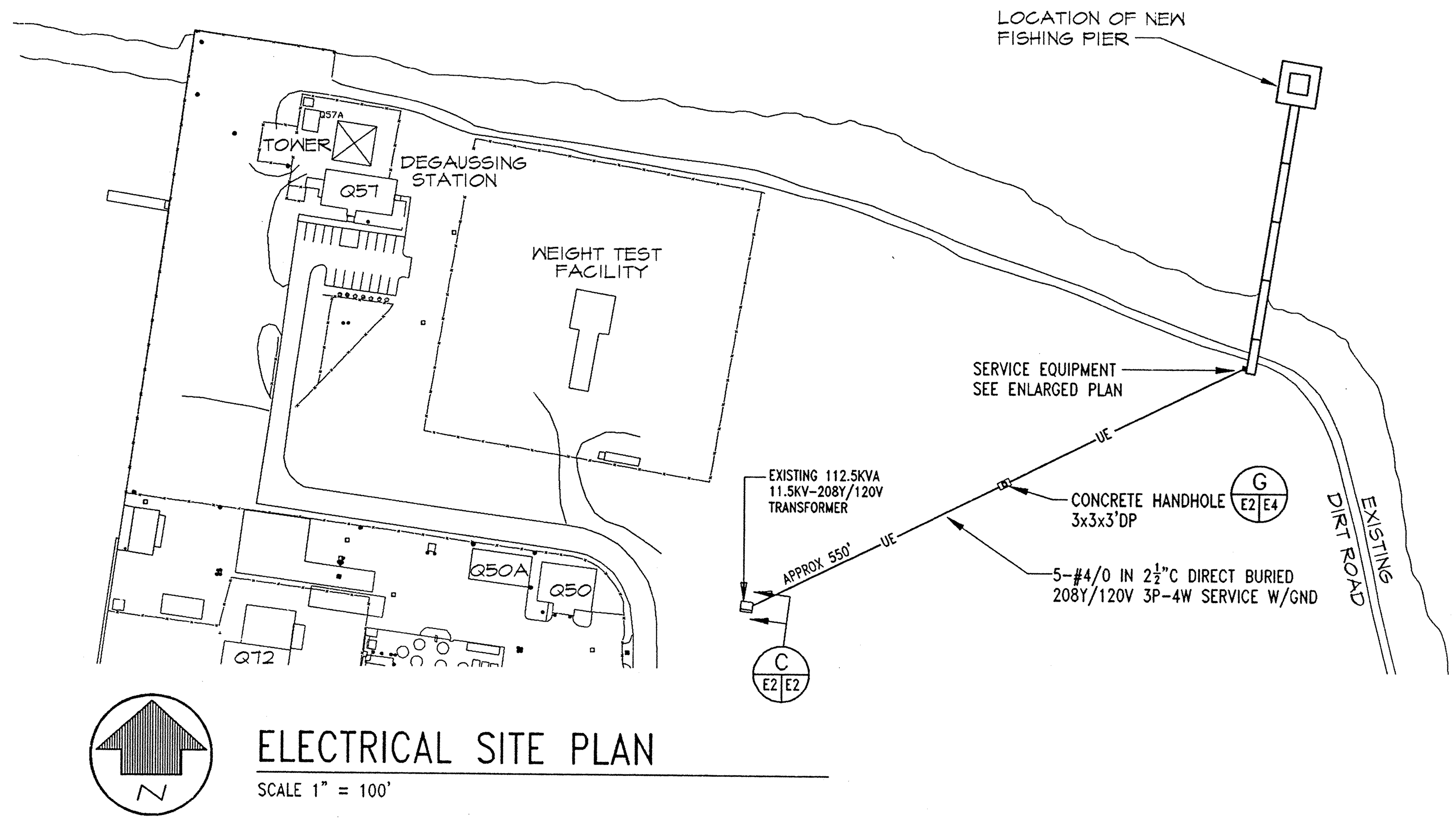


EQUIPMENT SUPPORT STRUCTURE

1. VERTICLE SUPPORT MEMBER – (2) 3-1/2 X 3-1/2" WELDED STEEL ANGLES SEE TOP VIEW
2. HORIZONTAL SUPPORT MEMBER – C4 STEEL CHANNEL WELDED TO VERTICAL SUPPORT MEMBERS
3. SET VERTICAL STEEL SUPPORT MEMBERS IN CONCRETE BASE AS INDICATED
4. LOCATE VERTICAL AND HORIZONTAL SUPPORTS COORDINATED WITH ENCLOSURE MOUNTING FEET.
5. CONCRETE BASE – 18 INCHES WIDE X 24 INCHES DEEP X LENGTH REQUIRED
6. PAINT STRUCTURAL MEMBERS PRIOR TO MOUNTING EQUIPMENT, (1) COAT RUST INHIBITING PRIMER AND (2) COATS GRAY ENAMEL PAINT.

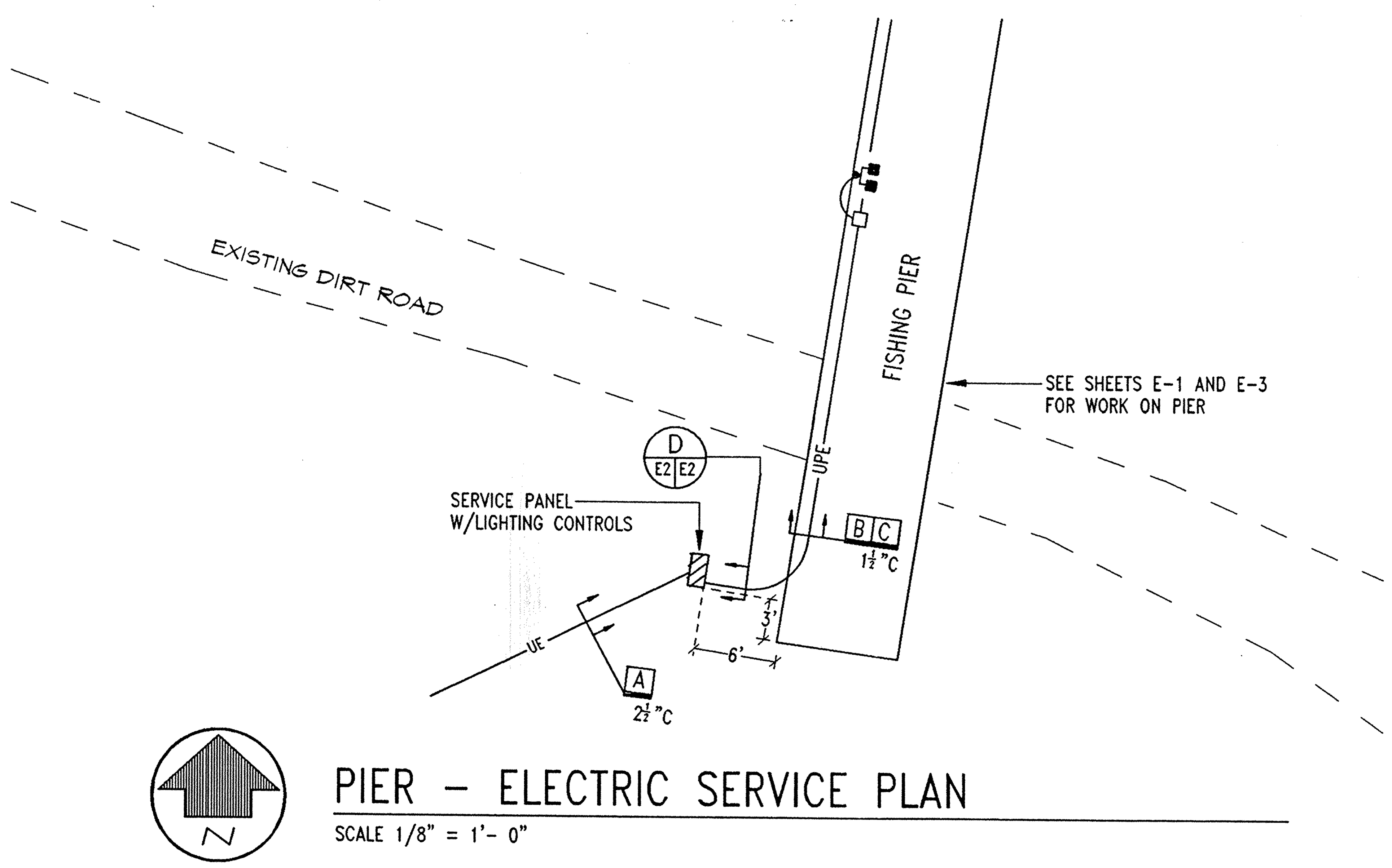
D E2 E2 DETAIL – SERVICE PANEL W/LIGHTING CONTROLS

SCALE 1" = 1' – 0"



ELECTRICAL SITE PLAN

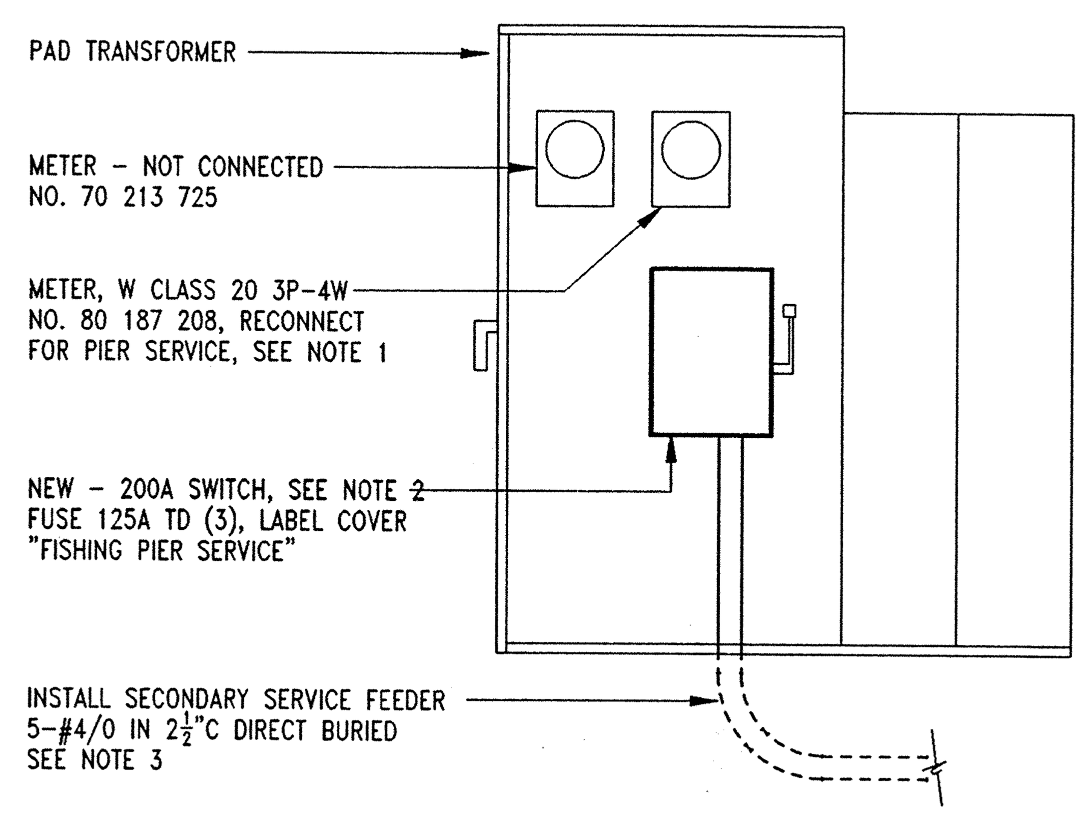
SCALE 1" = 100'



PIER – ELECTRIC SERVICE PLAN

SCALE 1/8" = 1'– 0"

ALL EQUIPMENT IS EXISTING UNLESS OTHERWISE NOTED



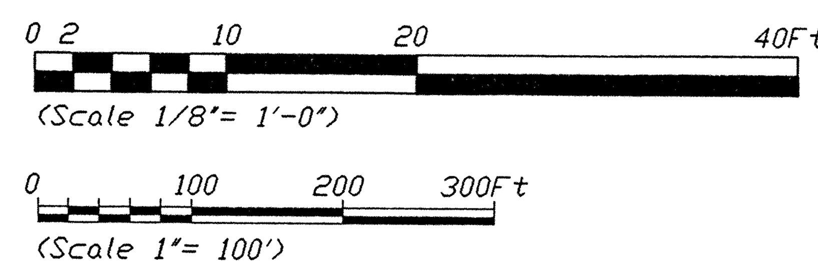
C E2 E2 TRANSFORMER – RIGHT SIDE ELEVATION

NOT TO SCALE

NOTES

1. PROVIDE ELECTRIC SERVICE TO PIER USING EXISTING EQUIPMENT AS INDICATED. MODIFY TRANSFORMER SECONDARY MAIN WIRING AND METER CONNECTIONS AS REQUIRED.
2. REMOVE EXISTING 2-WIRE 100A SWITCH AND INSTALL A 3-WIRE S/N 200A CLASS R FUSIBLE SWITCH IN NEMA 3R ENCLOSURE. PROVIDE FOUR #4/0 AND #4/0 EQUIPMENT GND CONNECTION TO TRANSFORMER SEC MAIN.
3. PROVIDE DIRECT BURIED UNDERGROUND SECONDARY SERVICE FEEDER AS INDICATED, MINIMUM 24" COVER, WITH BURIED WARNING TAPE 12" BELOW GRADE.

GRAPHIC SCALES



7/10/97 JBG(?)
BID ITEM NO 2 N.I.C.

SYM	DESCRIPTION	DATE	APPROVED

REVISIONS

JOB ORDER NO. 1646544
DES. KJF DR. KJF CHK. AMD
E.I.C. MKM
BR. MGR. D N ROGERS
DIV. DIR R. H. IVES, JR
APPROVED: DATE:
FOR C.O. COMMANDER, NAVFAC

SUBMITTED BY:
FIRM MEMBER / PRINCIPAL
APPROVED: DATE:
SANDRA HAWKINSON 6/3/96
SATISFACTORY TO ACTIVITY
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P W C NORVA

SEAL

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Q-AREA
ELECTRICAL SERVICE

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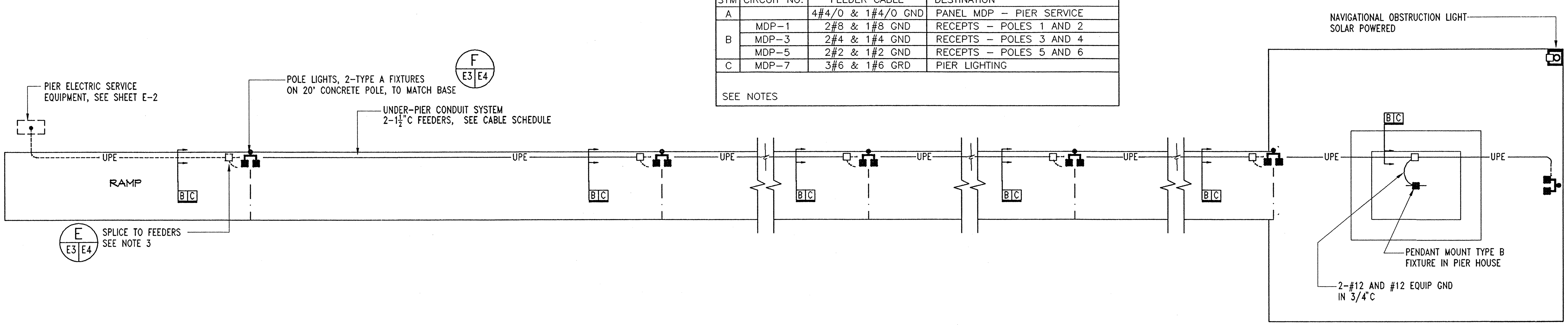
CONST. CONT. NO. N62470-96-B-7429
SPEC. 05-96-7429
NAVFAC DRAWING NO. 4333345
PWC DRAWING NO. 15,499J
SHEET 11 OF 13

E-2

CABLE SCHEDULE

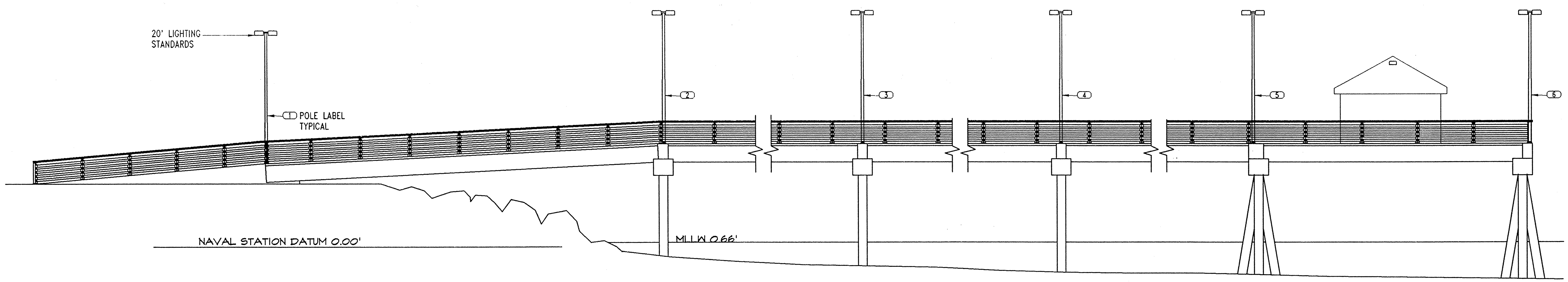
SYM	CIRCUIT NO.	FEEDER CABLE	DESTINATION
A		4#4/0 & 1#4/0 GND	PANEL MDP – PIER SERVICE
B	MDP–1	2#8 & 1#8 GND	RECEPTS – POLES 1 AND 2
	MDP–3	2#4 & 1#4 GND	RECEPTS – POLES 3 AND 4
	MDP–5	2#2 & 1#2 GND	RECEPTS – POLES 5 AND 6
C	MDP–7	3#6 & 1#6 GRD	PIER LIGHTING

SEE NOTES



PIER - ELECTRICAL PLAN

SCALE: 1/8" = 1'-0"



PIER PROFILE

SCALE: 1/8" = 1'-0"

NOTES

1. PROVIDE PIER ELECTRICAL SYSTEM CONSISTING OF 120/208V 3P-4W SERVICE AS SHOWN ON SHEET E-2, A SINGLE 3-PHASE 208V-20A LIGHTING CIRCUIT, AND THREE 120V-20A RECEPTACLE CIRCUITS. PIER ELECTRICAL INCLUDES ALL WIRING, POLES, LIGHT FIXTURES AND RECEPTACLES AS INDICATED. THE OBSTRUCTION LIGHT IS PROVIDED PER BASE BID.
2. PROVIDE PIER BRANCH CIRCUITS AS INDICATED IN THE CABLE SCHEDULE. ALL EQUIPMENT IS REFERENCED TO POLE NUMBERS SHOWN.
3. MAKE BRANCH CIRCUIT TAP CONNECTIONS IN JUNCTION BOXES PER DETAIL E WITH APPROVED INSULATED TAP CONNECTORS. ALL TAP CONDUCTORS SHALL BE SIZE #12AWG.
4. CONNECT POLE LIGHTS FOR BALANCED 3-PHASE LOAD BY CONNECTING FIXTURES TO PHASES AS FOLLOWS: POLES 1 AND 6 TO AB, POLES 2 AND 5 TO BC, POLES 3 AND 4 TO AC, AND PIER HOUSE LIGHT TO AB.

PANEL "MDP"

120/208V 3PH-4W
125A-3P MCB 10KAIC

LOAD SERVED	LOAD A	LOAD B	LOAD C	BKR TRIP	WIRE SIZE	CKT NO.	A B C	CKT NO.	WIRE SIZE	BKR TRIP	LOAD A	LOAD B	LOAD C	LOAD SERVED
RECEPT – POLES 1 & 2	16			20	8	1		2		20				SPARE
RECEPT – POLES 3 & 4		16		20	4	3		4		20				SPARE
RECEPT – POLES 5 & 6			16	20	2	5		6		20				SPARE
PIER LIGHTS	15				6	7		8						SPARE
		15		3P/20	6	9		10		3P/20				
			15		6	11		12						
SPACE						13		14						SPACE
						15		16						
						17		18						
TOTAL LOAD PER PHASE	31	31	31								0	0	0	

LOCATED AT HEAD OF FISHING PIER

LOAD (AMPS)
A = 31 B = 31 C = 31

FEEDER: 4/0 AWG
AIC EACH CB: 10,000

GRAPHIC SCALE

5' 0 5' 10' 20'

1/8" = 1'-0"

SYM	DESCRIPTION	DATE	APPROVED
	BID ITEM NO. 2 N.I.C.	7/10/97	FBG

REVISIONS

JOB ORDER NO. 1646544
DES. KJF DR. KJF CHK. AMD
E.I.C. MKM
BR. MGR. D. N. ROGERS
DIV. DIR R. H. IVES, JR
APPROVED: DATE: 5/21/96
FOR C.O. COMMANDER, NAVFAC

SUBMITTED BY:
FIRM MEMBER / PRINCIPAL
APPROVED: DATE:
SANDRA HAWKINSON 6/3/96
SATISFACTORY TO ACTIVITY
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PIER ELECTRICAL

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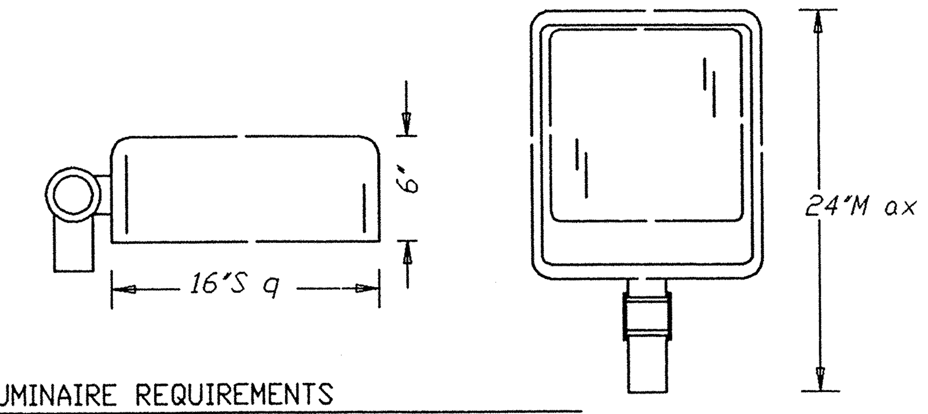
SHEET 12 OF 13

E-3



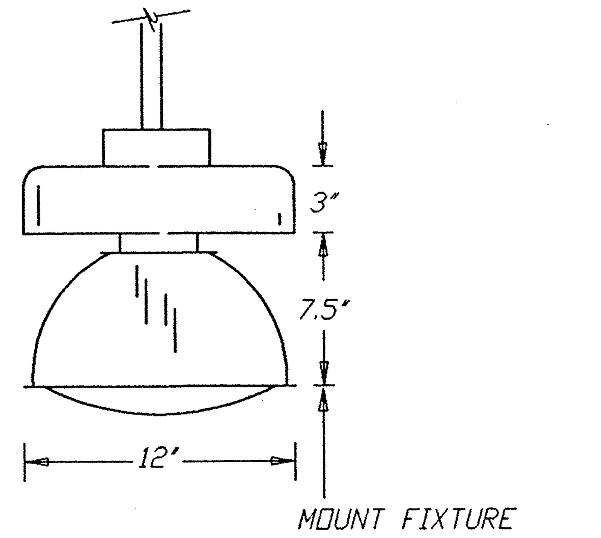
LUMINAIRE REQUIREMENTS

1. SEAMLESS DIE CAST ALUMINUM HOUSING WITH BLACK ACRYLIC POWDER PAINT FINISH (2-4 MILS THICK)
2. CLEAR TEMPERED GLASS LENS ON INTERNAL HINGED DIE CAST DOOR FRAME FOR EASY ACCESS TO BALLAST AND LAMP
3. FACTORY INSTALLED HIGH POWER FACTOR BALLAST CORE AND COIL CWA MULTI-TAP TYPE, 460W INPUT STARTING TEMP -20 DEG F, CBM CERTIFIED
4. OPTICS - FIELD ADJUSTABLE TYPE 2 SET FOR MAX SIDE DISTRIBUTION
5. SLIPFITTER POLE MOUNT, 2' ADJUSTABLE FITTER OR ADJUSTABLE ANGLE YOKE FOR WALL MOUNTING
6. EPA RATING OF 3.46 FOR TWO IN-LINE TENON MOUNT POLE CONFIGURATION, 0 DEGREE TILT
7. COATED METAL HALIDE MOGUL BASE LAMP RATED 400W 70CRI, 20,000HRS LIFE .75 MEAN DEPRECIATION
8. UL LISTED FOR WET LOCATIONS

TYPE A

PIER LIGHT

SKETCH DATE	JUNE 1996 /kjf	STYLE	XL-2B



MOUNT FIXTURE
12' ABOVE PIER DECK

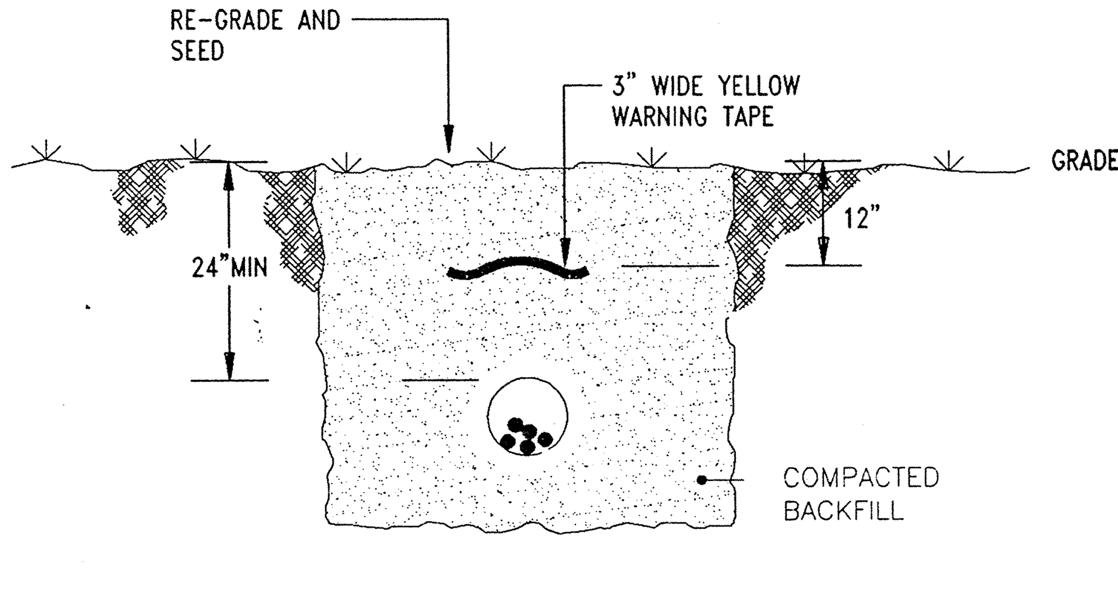
LUMINAIRE REQUIREMENTS

1. SEAMLESS DIE CAST ALUMINUM HOUSING WITH WHITE ACRYLIC POWDER PAINT FINISH (2-4 MILS THICK)
2. PRECISION INJECTION MOLDED ACRYLIC REFLECTOR WITH ETCHED AND ANODIZED ALUMINUM COLLAR
3. PROVIDE WITH CLEAR DOMED ACRYLIC LENS AND DUAL EXTERNAL FUSE FOR 208V CONNECTION
4. FACTORY INSTALLED HIGH POWER FACTOR BALLAST CORE AND COIL CWA MULTI-TAP TYPE, 200W INPUT STARTING TEMP -20 DEG F, CBM CERTIFIED
5. SLIDE SPLICE BOX FOR PENDENT MOUNTING VIA 1/2"C
6. CLEAR HIGH PRESSURE SODIUM LAMP RATED 150W 22CRI, 24,000HRS LIFE .90 MEAN DEPRECIATION
7. UL LISTED FOR WET LOCATIONS

TYPE B

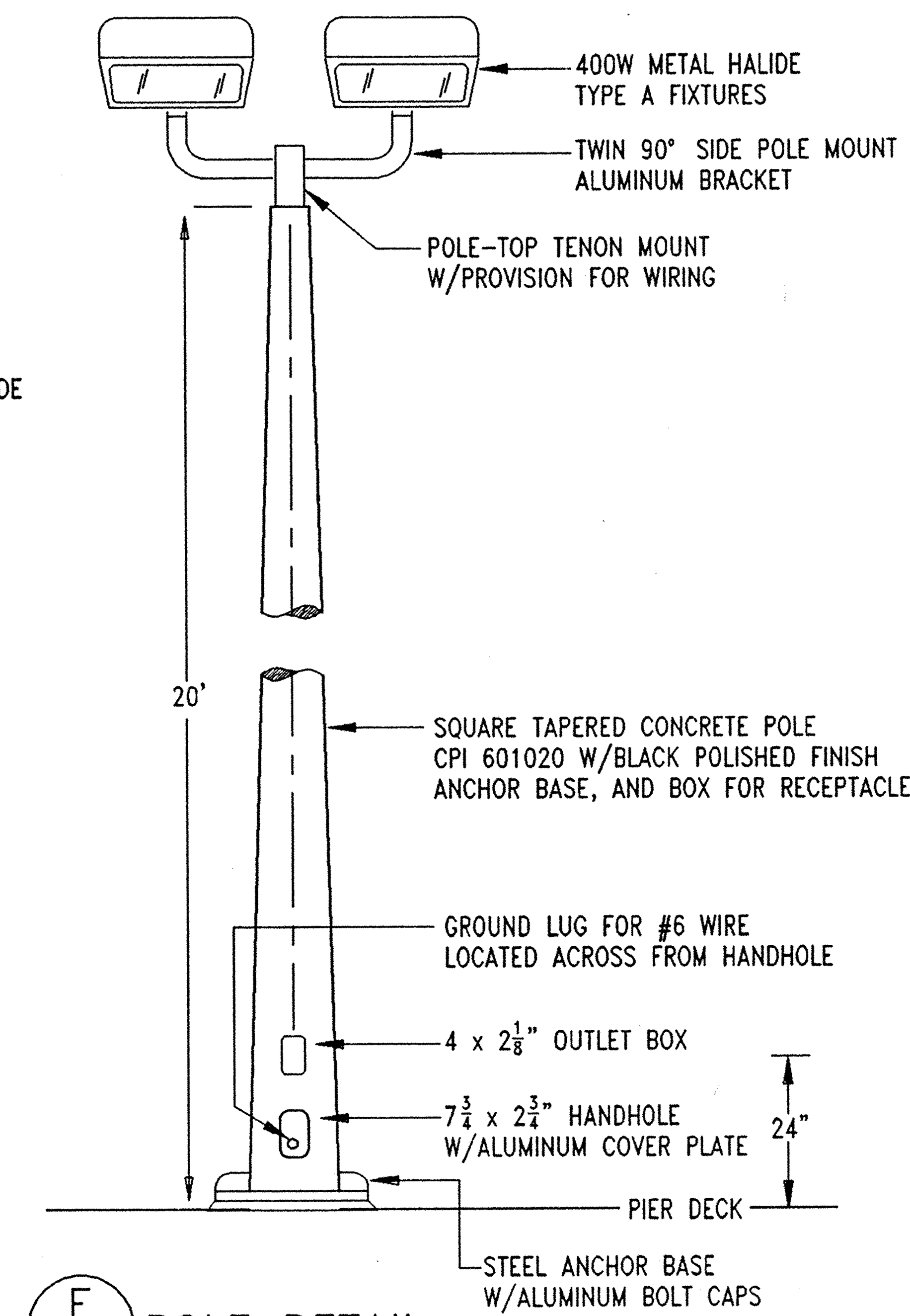
MINI PRISMATIC GAZEBO LIGHT

SKETCH DATE	JUNE 1996 /kjf	STYLE	XL-2C



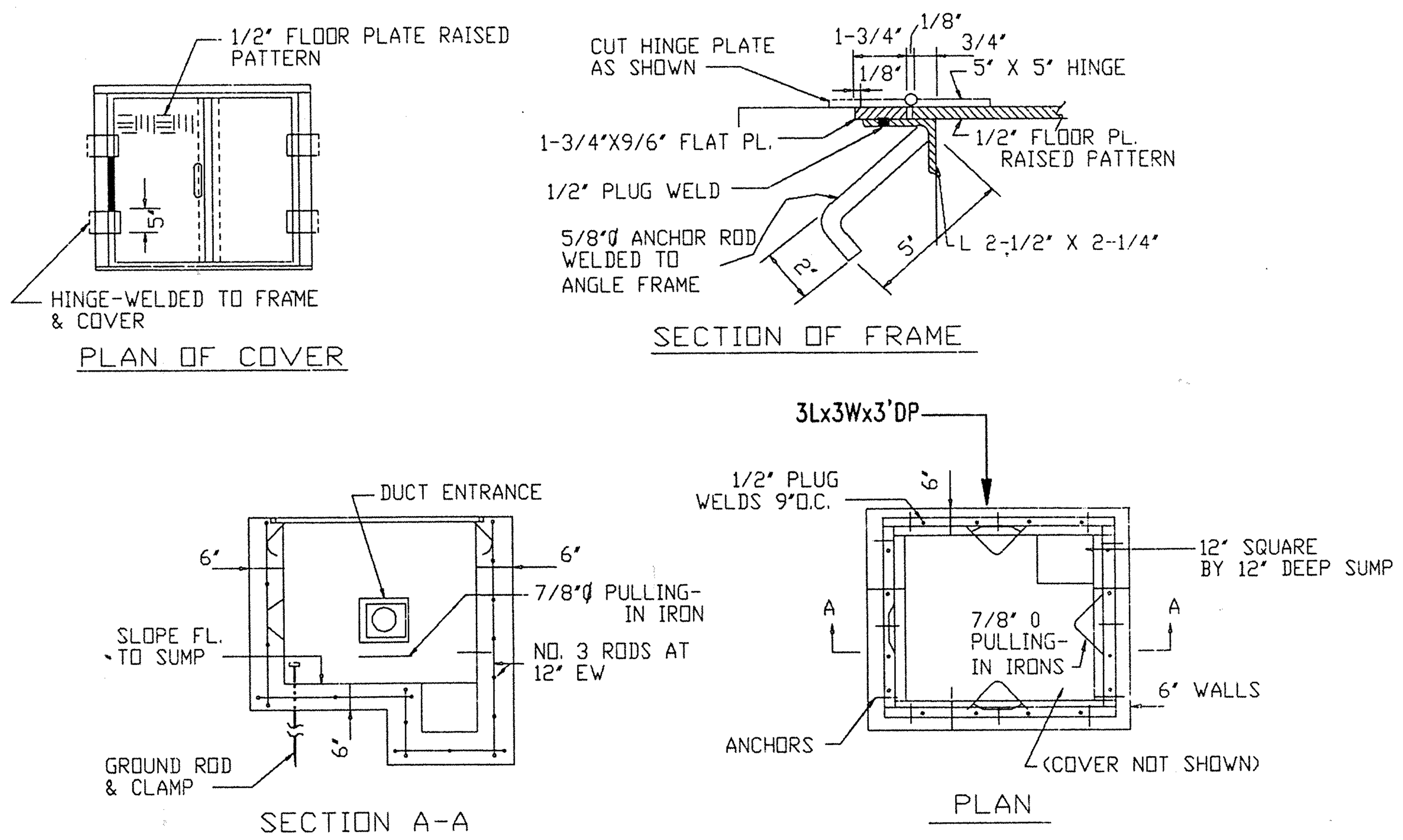
H (E1 E4) DIRECT BURIED CONDUIT

NO SCALE



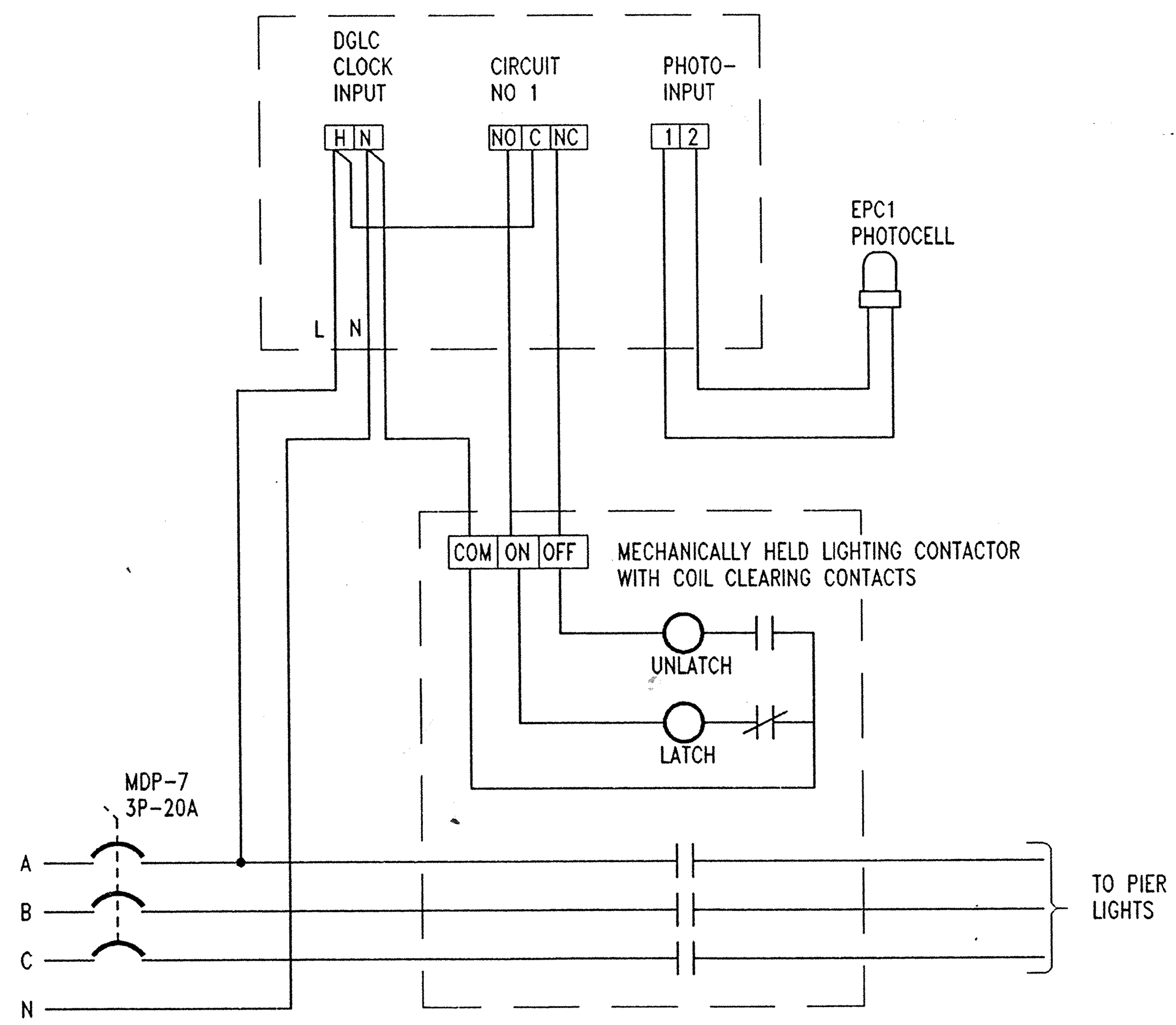
F (E3 E4) POLE DETAIL

NO SCALE



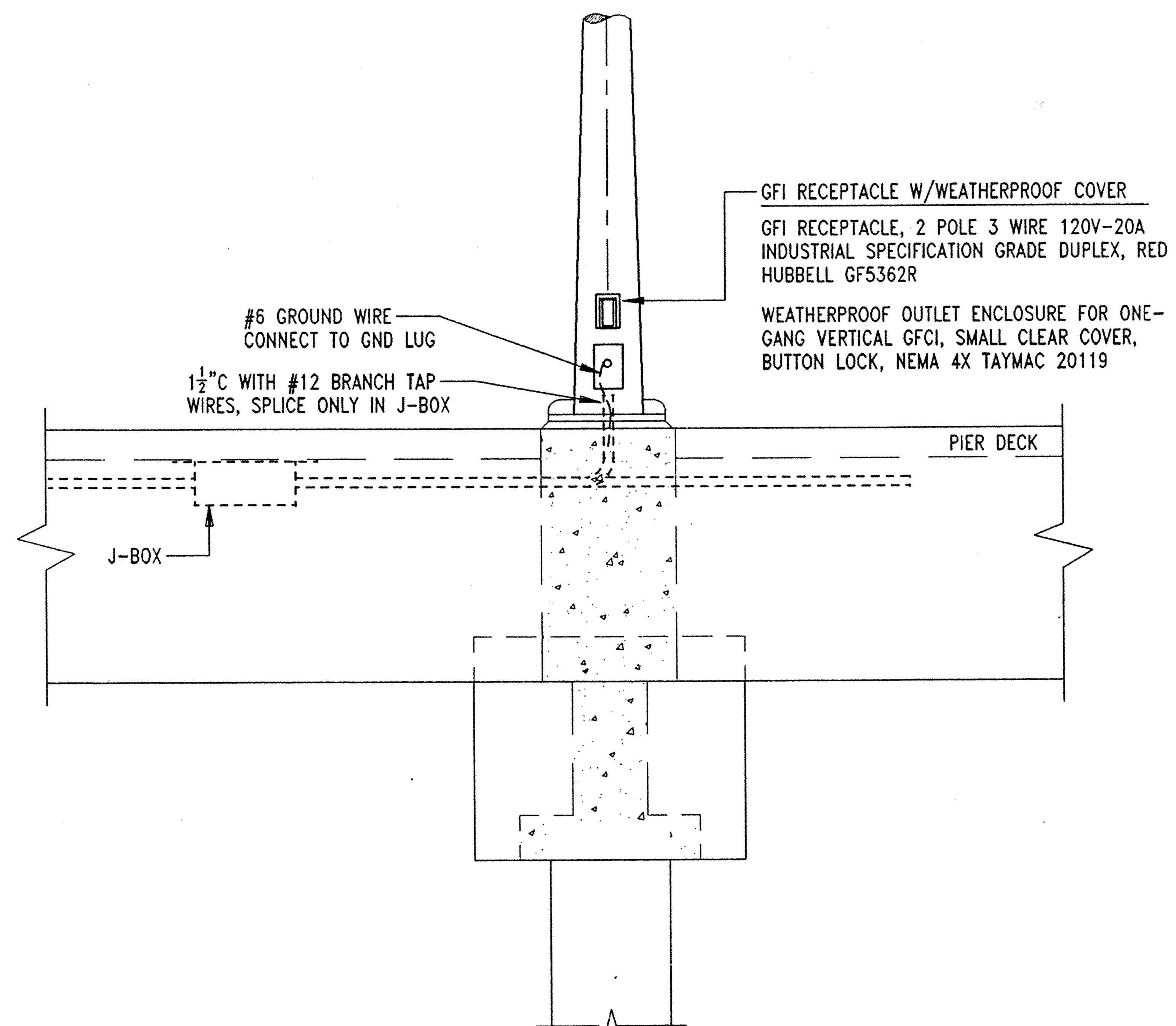
G (E2 E4) CONCRETE HANDHOLE

NO SCALE



LIGHTING CONTROL WIRING DIAGRAM

NO SCALE



E (E3 E4) POLE LIGHT INSTALLATION ON PIER

NO SCALE

SYM	DESCRIPTION	DATE	APPROVED
	BID ITEM NO. 2 NIC	7/10/97	[signature]

REVISIONS

SUBMITTED BY:
FIRM MEMBER / PRINCIPAL
APPROVED: SANDRA HAWKINSON DATE: 6/3/96
SATISFACTORY TO ACTIVITY
APPROVED: DATE:
SATISFACTORY TO COMNAVBASE

JOB ORDER NO. 1646544
DES. KJF DR. KJF CHK. AMD
E.I.C. MKM
BR. MGR. D. N. ROGERS
DIV. DIR R. H. IVES, JR
APPROVED: [signature] DATE: 6/27/96
FOR C.O. COMMANDER, NAVFAC

P W C NORVA

SEAL

DEPT. OF THE NAVY NAVFACENGCOM
NAVY PUBLIC WORKS CENTER
NAVAL STATION NORFOLK, VIRGINIA
NAVAL STATION NORFOLK, VIRGINIA

FISHING PIER
Q-AREA

ELECTRICAL DETAILS

CODE ID. NO. 80091
DRAWING SIZE: D
CONST. CONT. NO. N62470-96-B-7429
SPEC. 05-96-7429
NAVFAC DRAWING NO. 4333347
PWC DRAWING NO. 15,499L
SHEET 13 OF 13

E-4